Monetary Policy, Competition Policy, and the Dynamics of Market Power

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Abstract

We develop a two-sector New Keynesian model in which firms allocate scarce organizational resources between productive capability and customer capital. Customer capital lowers demand elasticity and raises desired markups, monetary policy therefore affects not only aggregate demand but also incentives to build market power. Disciplined by U.S. firm-level accounting data and business-cycle moments, the model implies that endogenous demand-to-market-power transmission accounts for about 78 percent of markup variance. Firm-level local projections support the mechanism: customer-capital-intensive firms increase market-building after demand expansions and monetary easing. Implementable-rule frontiers show that competition policy, by targeting customer acquisition, complements monetary stabilization and raises welfare.

Keywords: market power, customer capital, monetary policy, competition policy, markups

JEL Classification: E31, E52, L11, L40, L51

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1 Introduction

Two policy debates bear on the same phenomenon but rarely speak to each other. Monetary economics studies how interest rates move output, inflation, and factor utilization, while taking firms' desired markups as exogenous primitives or as reduced-form cost-push wedges orthogonal to firms' own decisions (Woodford, 2003; Galí, 2015). Competition policy, revived by evidence that U.S. product markets have grown less competitive—rising aggregate markups and concentration, a falling labor share, and a shift of activity toward large, high-markup firms (De Loecker et al., 2020; Autor et al., 2020; Gutiérrez and Philippon, 2017; Covarrubias et al., 2020; Grullon et al., 2019; Aghion et al., 2023)—asks how to restrain that market power, but treats the macroeconomy as a backdrop. Both debates abstract from a fact the same evidence stresses: firms actively invest in demand-side advantages—brands, customer relationships, distribution, and selling capacity—rather than in productive efficiency alone (Syverson, 2019; Eeckhout, 2021). Once market power is something firms build, two questions follow that neither literature answers in isolation: does monetary policy shape the accumulation of market power, and how does it interact with competition policy aimed at the same margin?

This paper develops a tractable framework that addresses both. Firms own two slow-moving intangible stocks. The first, productive capability, raises efficiency. The second, customer capital, lowers the elasticity of demand the firm faces and so raises its desired markup, following the customer-markets tradition in which sales depend on a stock of attached customers (Gourio and Rudanko, 2014; Ravn et al., 2006). Building either stock draws on the same scarce pool of organizational resources (Eisfeldt and Papanikolaou, 2013; Corrado et al., 2009), so the firm continually allocates effort between productive-capability building and market-building. That single allocation margin is what both policies act on. Monetary policy moves it indirectly: by changing discount factors, current demand, the cyclical value of future customers, and the financing cost of customer acquisition (Gilchrist et al., 2017), it shifts the private return to market-building relative to productive-capability building. Competition policy moves it directly, by changing the durability of customer capital and the private return to acquiring customers. Because the two instruments operate on the same dynamic margin, their effects are not separable, and the natural object of study is their joint design.

We embed this mechanism in an otherwise standard two-sector New Keynesian economy. Prices and wages are sticky as in Rotemberg (1982) and Erceg et al. (2000); households accumulate physical capital subject to adjustment costs and form habits; and a strategic, customer-capital-intensive sector coexists with a competitive outside sector

under constant-elasticity final demand. Within the strategic sector the firm allocates organizational resources between productive capability and market-building, and a long-duration financing wedge makes customer-acquisition outlays sensitive to monetary innovations. Policy is set by two rules that lean on the allocation margin from opposite sides: an inertial Taylor rule for the nominal interest rate, and a competition-policy block—a tax on market-building and a depreciation instrument for customer capital—that responds to the cyclical environment. A natural, flexible-price output gap enters both the firm's valuation of customer capital and the competition-policy rule, where it measures the cyclical value of market-building relative to the efficient allocation.

The paper makes three contributions. First, it provides a sticky-price general-equilibrium model of endogenous market-power accumulation in which the central margin is the firm's internal allocation between productive capability and customer capital, rather than firm entry or exogenous markup wedges. Second, it shows that monetary and competition policy both act on this margin and analyzes their joint design, characterizing when the two instruments are complements. Third, it disciplines the new margins with U.S. firm-level accounting data and business-cycle moments and tests the model's distinctive cross-sectional prediction directly in the firm data.

The quantitative message is structural. In the calibrated model, business-cycle variation in markups is driven primarily by the endogenous response of firms' market-building decisions to demand and monetary conditions, not by exogenous markup shocks: demand disturbances account for the bulk of the variance of the aggregate markup. Demand expansions are the dominant and most persistent force raising customer-capital accumulation, while monetary innovations move the same margin more indirectly, through the policy-rate, demand, inflation, and financing blocks. We corroborate the central prediction in a firm-year panel: in local projections that interact aggregate conditions with firms' pre-period market-building intensity, customer-capital-intensive firms robustly raise market-building after aggregate demand expansions. Consistent with the structural mechanism, the firm-level monetary cross-section is weaker and not robustly identified at annual frequency, we therefore treat the demand channel as the firm-level evidence that disciplines the customer-capital mechanism.

Because market power here accumulates endogenously and responds to aggregate conditions, it is something policy can lean against rather than a fixed distortion, and the two instruments do so from opposite sides of the same margin. Competition policy acts on it directly: in the benchmark, a temporary market-building tax lowers customer acquisition, shifts organizational resources toward productive-capability building, and reduces the strategic-sector markup, with only limited movement in

the policy rate. Monetary policy reaches the same margin indirectly, thus the two are naturally studied together. Optimizing over an implementable class of rules, we find that they are complementary: the best joint rule raises welfare by more than either instrument alone, combining tighter nominal stabilization with a tax response to cyclical customer acquisition. The calibrated model matches the sign and approximate magnitude of output, inflation and output, interest-rate co-movement and of inflation volatility; the remaining quantitative tension is between aggregate output amplification and labor-share volatility.

The remainder of the paper is organized as follows. Section 2 relates the paper to the literature. Section 3 presents the model environment. Section 4 characterizes the equilibrium and states the analytical results used to interpret the simulations. Section 5 explains the quantitative discipline. Section 7 reports the positive results. Section 8 studies the joint policy problem. Section 9 concludes. The appendices collect the firm-level robustness analysis, the equilibrium system, and the proofs.

2 Related Literature

A long tradition models demand as depending on a stock of attached customers that firms build through past sales and marketing. Gourio and Rudanko (2014) formalize customer capital as a state accumulated through costly customer search and show that it produces sluggish, forward-looking pricing and amplifies cyclical movements in the labor wedge. Ravn et al. (2006) obtain a related demand-side state through "deep habits", which makes the price elasticity of demand depend on a good-specific stock and gives firms an intertemporal pricing motive. Gilchrist et al. (2017) add financial frictions to customer markets and show, in U.S. micro data and a calibrated model, that liquidity-constrained firms raised prices during the Great Recession to harvest existing customers: direct evidence that the financing cost of customer acquisition shapes price-setting. We build on these mechanisms but pose a different question: instead of studying pricing given customer capital, we treat customer-capital building as one of two competing intangible investments and characterize how monetary and competition policy reallocate the firm between them. The financing channel of Gilchrist et al. (2017) enters our model as a wedge on customer-acquisition outlays.

A growing literature shows that monetary policy is not neutral with respect to market power. Baqaee et al. (2024) argue that, because high-markup firms face more elastic demand and adjust quantities more, monetary policy has first-order supply-side effects, raising measured productivity when it stimulates these firms; Meier and Reinelt (2024) document that contractionary monetary shocks raise markup dispersion and lower ag-

gregate TFP, and rationalize this with sticky-price firms whose relative markups move with policy. Wang and Werning (2022) show how dynamic oligopoly reshapes the Phillips curve and the transmission of monetary shocks, and Aquilante et al. (2019) provide cross-country evidence that market power conditions monetary transmission. In monetary economies with decentralized trade, Choi and Rocheteau (2024) micro-found producer market power, and Kala et al. (2026) study how rent-seeking effort and market power respond dynamically to the interest rate. These papers take the level or distribution of market power as the object that monetary policy moves. We differ by making the firm's investment in future market power endogenous: monetary policy here changes not only current markups but the rate at which firms accumulate the customer capital that sustains them, which is what links monetary conditions to the medium-run path of market power.

Our motivating facts come from the literature measuring the secular rise in U.S. market power. De Loecker et al. (2020) estimate that aggregate markups rose from roughly 1.2 in 1980 to about 1.6 by 2016 and trace the macroeconomic implications; Autor et al. (2020) link the falling labor share to a reallocation of activity toward low-labor-share "superstar" firms; and Gutiérrez and Philippon (2017), Covarrubias et al. (2020), and Grullon et al. (2019) document rising concentration and weak investment. Aghion et al. (2023) provide a theory in which the spread of efficient, high-markup firms raises concentration and rents, and Edmond et al. (2023) quantify the welfare cost of the resulting markup dispersion, while Syverson (2019) and Eeckhout (2021) survey the debate. This work is largely about low-frequency trends and their welfare costs. We take the rise in market power as motivation rather than as our explanandum and contribute a business-cycle mechanism: ordinary demand and monetary fluctuations move firms' market-building investment, thus the accumulation of market power has a cyclical component that policy can influence.

On the nominal side we use the price-adjustment-cost formulation of Rotemberg (1982) and the staggered wage and price contracts of Erceg et al. (2000), with the standard New Keynesian links to Galí and Gertler (1999), Nekarda and Ramey (2020), Woodford (2003), and Galí (2015). For policy we follow the implementable-rule approach of Schmitt-Grohé and Uribe (2007) and the linear-quadratic welfare analysis around a distorted steady state of Benigno and Woodford (2005). Relative to this literature, our policy problem contains a second instrument competition policy, that enters the same dynamic margin as the interest rate, the design problem therefore is genuinely joint: the optimal monetary rule depends on how aggressively competition policy leans against customer acquisition, and conversely.

Finally, the paper relates to macro models with endogenous market structure. Bilbiie et al. (2012) make the number of products and the markup respond to entry

over the cycle, and Colciago and Etro (2010), Etro and Colciago (2010), and Jaimovich (2007) study how endogenous competition and firm dynamics generate countercyclical markups. In these models market power varies through the extensive margin of entry; in ours it varies through an intensive margin, incumbents' investment in customer capital, whose measurement we connect to the intangible- and organization-capital literature (Corrado et al., 2009; Eisfeldt and Papanikolaou, 2013). The distinction matters for policy, because a tax on customer-building acts on a margin that entry-based models do not contain.

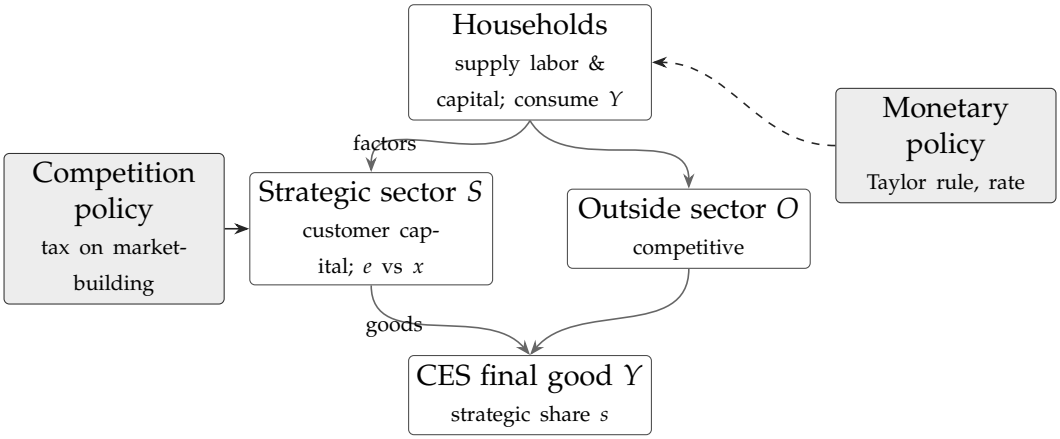
Taken together, these literature establish that market power has risen, that customer capital makes demand intertemporal, that monetary policy is not neutral with respect to markups, and that simple rules can be ranked by welfare. What none provides is a framework in which firms invest in market power along a margin that both monetary and competition policy move, disciplined by firm-level data and used to design the two policies jointly. Filling that gap is the major contribution of this paper.

3 Model

The economy has a representative household, a strategic sector with endogenous customer capital and sticky prices, a competitive outside sector, and monetary and competition-policy authorities. The strategic sector is disciplined by firm-level evidence on its steady-state revenue share. The central real friction is scarce organizational throughput: firms can build productive capability and customer capital, but both activities draw on internal resources whose marginal cost rises with total throughput and falls with inherited organizational capital. The model also includes sticky wages and a private market-power disturbance that changes how efficiently customer capital lowers demand elasticity. A long-duration financing block exposes market-building costs to monetary innovations without introducing an additional independent shock. Figure 1 summarizes the model's structure.

At the beginning of period t, firms inherit sectoral capital $(K_{S,t-1}, K_{O,t-1})$, productive capability $\overline{A}_{t-1}$, customer capital $\bar{m}_{t-1}$, and organizational capital H_{t-1}. Shocks are realized, policy rules set $(R_t, \delta_{m,t}, \tau_{r,t})$, and firms choose sectoral production, prices, productive capability building x_t, market-building effort e_t, and organizational investment i_t^H. Productive capability is implemented within the period and enters current strategic production. Customer capital is slower moving: current market-building affects current demand directly, but becomes inherited customer capital with a one-period delay.

Figure 1: Model structure



Note: Households supply labor and capital and consume the final good Y, a constant-elasticity aggregate of a strategic, customer-capital-intensive sector (S) and a competitive outside sector (O). Competition policy acts directly on the strategic sector's market-building; monetary policy operates through the nominal block.

3.1 Households

The representative household solves

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\exp(z_t^B) \frac{(C_t - hC_{t-1})^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\varphi}}{1+\varphi} \right]. \tag{1}$$

Here $\beta \in (0,1)$ is the discount factor, $h \in [0,1)$ the degree of external habit, $\sigma > 0$ the inverse intertemporal elasticity of substitution, $\varphi > 0$ the inverse Frisch elasticity of labor supply, and $\chi > 0$ the weight on the disutility of hours N_t; C_t is consumption. The preference shifter z_t^B is an aggregate-demand shock that follows $z_t^B = \rho_B z_{t-1}^B + \varepsilon_t^B$ and is the model's main business-cycle disturbance. The marginal utility object used in the Euler equation is

$$\lambda_t = \exp(z_t^B)(C_t - hC_{t-1})^{-\sigma} - \beta h \, \mathbb{E}_t \left[\exp(z_{t+1}^B)(C_{t+1} - hC_t)^{-\sigma} \right], \tag{2}$$

and the household first-order conditions (F.O.C) are

$$1 = \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{R_t}{\pi_{t+1}} \right], \tag{3}$$

$$\pi_{w,t} = \pi_t w_t / w_{t-1}, \tag{4}$$

$$\theta_w \pi_{w,t}(\pi_{w,t} - 1) = \varepsilon_w \frac{\chi N_t^{\varphi}}{\lambda_t w_t} - (\varepsilon_w - 1)\exp(z_t^W) + \theta_w \beta \mathbb{E}_t \left[\frac{\lambda_{t+1}}{\lambda_t} \pi_{w,t+1}(\pi_{w,t+1} - 1) \frac{w_{t+1} N_{t+1}}{w_t N_t} \right]. \tag{5}$$

Equation (3) is the consumption Euler equation, in which R_t is the gross nominal interest rate and π_{t+1} gross price inflation. Households supply differentiated labor through a union that sets the nominal wage subject to a quadratic (Rotemberg) adjustment cost, which yields the wage Phillips curve (5): here w_t is the real wage, $\pi_{w,t}$ gross wage inflation, $\theta_w \geq 0$ the degree of wage stickiness, and $\varepsilon_w > 1$ the elasticity of substitution across labor varieties, so that $\varepsilon_w/(\varepsilon_w - 1)$ is the desired wage markup. The wage-markup shock follows $z_t^W = \rho_W z_{t-1}^W + \varepsilon_t^W$. The real discount factor used by firms is

$$\text{disc}_{t,t+1} \equiv \frac{\pi_{t+1}}{R_t}. \tag{6}$$

3.2 Two-sector production and final demand

The final good is a CES aggregate of strategic output $Y_{S,t}$ and outside output $Y_{O,t}$. The strategic-sector demand weight is

$$\Omega_{S,t} = \left[1 + \exp\left\{-\left(\log\frac{\Omega_S}{1-\Omega_S} + \phi_{\Omega B} z_t^B + \phi_{\Omega M} z_t^M\right)\right\}\right]^{-1}. \tag{7}$$

In the steady-state solver, Ω_S and outside productivity are backed out so that the strategic-sector revenue share equals the firm-level target $s_S^{ss} = 0.302$. Away from steady state, demand shocks move sectoral composition through $\phi_{\Omega B}$. The auxiliary coefficient $\phi_{\Omega M}$ lets autonomous market-power disturbances move sectoral composition directly; the benchmark sets $\phi_{\Omega M} = 0$.

Final-good demands and the price index are

$$Y_{S,t} = \Omega_{S,t} p_{S,t}^{-\eta_Y} Y_t, \tag{8}$$

$$Y_{O,t} = (1 - \Omega_{S,t}) p_{O,t}^{-\eta_Y} Y_t, \tag{9}$$

$$1 = \Omega_{S,t} p_{S,t}^{1-\eta_Y} + (1 - \Omega_{S,t}) p_{O,t}^{1-\eta_Y}. \tag{10}$$

The final good Y_t is the numeraire; $p_{S,t}$ and $p_{O,t}$ are the relative prices of the strategic and outside goods, and $\eta_Y > 1$ is the elasticity of substitution between them.

The strategic sector has CES production,

$$G_{S,t} = \alpha K_{S,t-1}^{\rho_{KN}} + (1-\alpha)(N_{S,t} - \bar{N})^{\rho_{KN}}, \tag{11}$$

$$Y_{S,t} = \exp(z_t^A)\overline{A}_t G_{S,t}^{1/\rho_{KN}}, \tag{12}$$

Strategic output combines predetermined capital $K_{S,t-1}$ and labor $N_{S,t}$ through the CES aggregator $G_{S,t}$ with substitution parameter ρ_{KN} (elasticity of substitution $1/(1-$

ρ_{KN})), where $\bar{N}$ is a fixed overhead-labor requirement. Productivity has a neutral total-factor-productivity component z_t^A (with $z_t^A = \rho_A z_{t-1}^A + \varepsilon_t^A$) and the endogenous Hicks-neutral productive-capability stock $\overline{A}_t$ accumulated from past building effort x_t. Cost minimization gives the factor prices, in which mc_t is the strategic sector's real marginal cost and $r_{S,t}^k$ the rental rate on strategic capital:

$$w_t = p_{S,t}(1-\alpha)mc_t Y_{S,t}^{1-\rho_{KN}}(\exp(z_t^A)\overline{A}_t)^{\rho_{KN}}(N_{S,t}-\bar{N})^{\rho_{KN}-1}, \tag{13}$$

$$r_{S,t}^k = p_{S,t}\alpha mc_t Y_{S,t}^{1-\rho_{KN}}(\exp(z_t^A)\overline{A}_t)^{\rho_{KN}}K_{S,t-1}^{\rho_{KN}-1}. \tag{14}$$

The outside sector is perfectly competitive, with Cobb–Douglas technology, capital share α_O, and productivity A_O; labor is mobile across sectors so a single real wage w_t clears the labor market:

$$Y_{O,t} = A_O K_{O,t-1}^{\alpha_O} N_{O,t}^{1-\alpha_O}, \tag{15}$$

$$w_t = p_{O,t}(1-\alpha_O)Y_{O,t}/N_{O,t}, \tag{16}$$

$$r_{O,t}^k = p_{O,t}\alpha_O Y_{O,t}/K_{O,t-1}, \tag{17}$$

$$N_t = N_{S,t} + N_{O,t}. \tag{18}$$

The economy-wide markup μ_t^{agg} (final output divided by total variable cost) and the aggregate labor share LS_t are measured as

$$\mu_t^{agg} = \frac{Y_t}{mc_t p_{S,t}Y_{S,t} + p_{O,t}Y_{O,t}}, \tag{19}$$

$$LS_t = \frac{w_t N_t}{Y_t}. \tag{20}$$

3.3 Customer capital, markups, and pricing

Strategic-sector desired markups are governed by a customer-capital elasticity:

$$d_t = 1 + \bar{m}_t + \zeta_e(e_t - E^{ss}), \tag{21}$$

$$\varepsilon_t = \bar{\varepsilon} - \phi_m \exp(\phi_{zM} z_t^M)\log d_t, \tag{22}$$

$$\mu_t^* = \frac{\varepsilon_t}{\varepsilon_t - 1}. \tag{23}$$

Here d_t is the demand-elasticity index: it rises with the inherited customer-capital stock $\bar{m}_t$ and with current market-building e_t measured relative to its steady-state level E^{ss}, where $\zeta_e \geq 0$ scales the latter channel. The baseline elasticity is $\bar{\varepsilon}$, and $\phi_m > 0$ governs how strongly a larger customer base compresses the perceived elasticity ε_t and hence raises the desired gross markup $\mu_t^* = \varepsilon_t/(\varepsilon_t - 1)$. Customer capital persistently lowers demand elasticity. Current market-building shifts demand conditions

around the steady state. The autonomous market-power shock z_t^M changes the efficiency with which customer capital translates into lower demand elasticity. When z_t^M is high, a given customer-capital stock produces a larger reduction in demand elasticity and hence a higher desired markup.

Strategic-sector firms set prices subject to a quadratic (Rotemberg) price-adjustment cost governed by the stickiness parameter θ. With $\pi_{S,t} = \pi_t p_{S,t}/p_{S,t-1}$ denoting gross strategic-sector inflation, the price-setting equation is

$$\theta\pi_{S,t}(\pi_{S,t}-1) = 1-\varepsilon_t+\varepsilon_t mc_t+\phi_P z_t^P+\theta\mathbb{E}_t\left[\text{disc}_{t,t+1}\frac{Y_{S,t+1}}{Y_{S,t}}\pi_{S,t+1}(\pi_{S,t+1}-1)\right]. \tag{24}$$

The forcing term $1-\varepsilon_t+\varepsilon_t mc_t$ is the standard Rotemberg expression with a time-varying desired markup; absent adjustment costs it would set the realized markup to its desired level, $mc_t = 1/\mu_t^*$. The disturbance z_t^P, with loading ϕ_P, is a price-markup wedge that shifts inflation conditional on marginal cost and desired markups and is absent from the flexible-price natural allocation.

3.4 Organizational capacity and intangible investment

The model uses inherited organizational capital as a quasi-fixed factor that lowers the cost of current organizational throughput. Define

$$T_t = x_t + \omega_e e_t, \tag{25}$$

where T_t is total organizational throughput in period t, x_t is the productive-capability building effort or investment flow, e_t is market-building effort, and $\omega_e > 0$ is the organizational intensity of market-building relative to productive-capability building. Thus one unit of x_t uses one unit of organizational throughput, while one unit of e_t uses ω_e units. The resource cost of producing throughput is

$$\Gamma_t = \frac{\phi_O}{1+\xi_O}T_t^{1+\xi_O}H_{t-1}^{-\psi_O}, \qquad \xi_O > 0,\ \psi_O > 0. \tag{26}$$

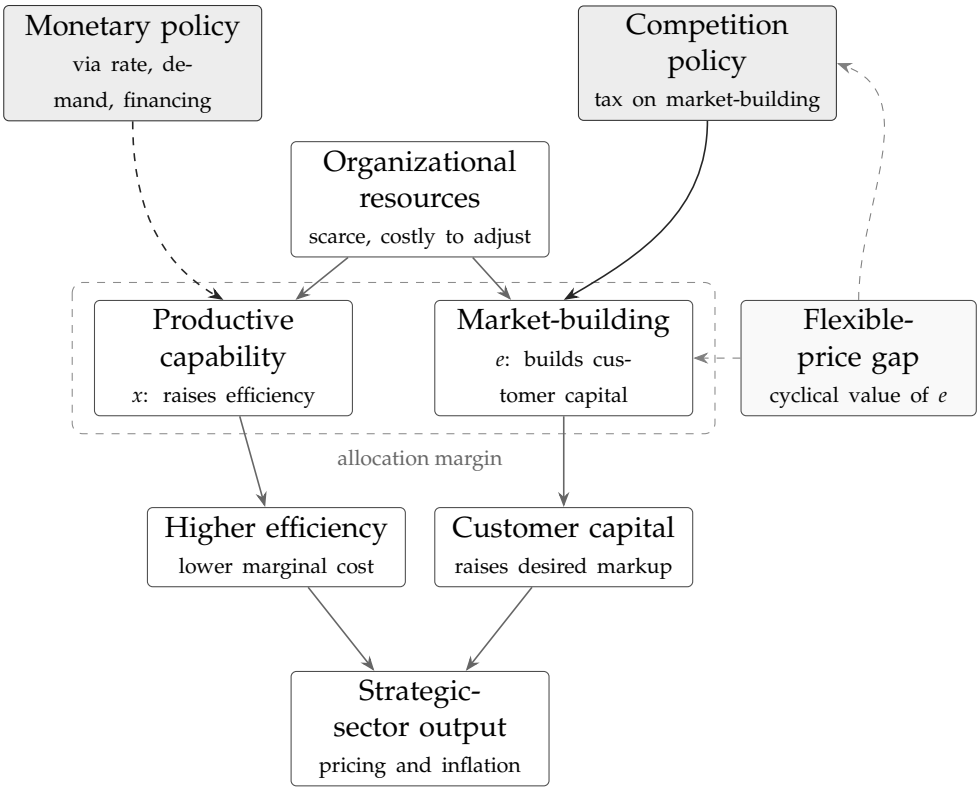
Its marginal cost is therefore

$$\nu_t^H = \frac{\partial\Gamma_t}{\partial T_t} = \phi_O T_t^{\xi_O}H_{t-1}^{-\psi_O}. \tag{27}$$

This allocation between x_t and e_t, and how the two policy instruments bear on it, is summarized in Figure 2.

The scale $\phi_O > 0$ and curvature $\xi_O > 0$ determine how steeply the marginal cost of

Figure 2: The firm's allocation margin



Note: The strategic firm splits scarce organizational resources between productive-capability building (x) and market-building (e). Monetary policy shifts the relative return indirectly (dashed); competition policy taxes market-building directly (solid); a flexible-price output gap informs the value of market-building and the competition-policy rule. Market-building accumulates customer capital and raises the desired markup, while productive capability raises efficiency—both feeding strategic-sector pricing and output.

throughput rises with the total volume of intangible activity, while $\psi_O > 0$ measures how strongly inherited organizational capital H_{t-1} lowers that cost. Because H_{t-1} no longer imposes the accounting identity $dx_t + \omega_e de_t = 0$ on impact, productive capability building and market-building are true joint choices disciplined by a convex marginal cost.

Organizational capacity evolves according to

$$H_{t+1} = (1 - \delta_H)H_t + i_t^H. \tag{28}$$

Productive capability and customer capital evolve as

$$\mathcal{S}_t^A = \left(\frac{Y_{S,t}}{Y_S^{ss}}\right)^{\phi_{AX}}, \tag{29}$$

$$A_t^{build} = \eta_A x_t^{\xi_x} \mathcal{S}_t^A, \tag{30}$$

$$\overline{A}_t = (1 - \delta_A)\overline{A}_{t-1} + A_t^{build}, \tag{31}$$

$$\bar{m}_t = (1 - \delta_{m,t-1})\bar{m}_{t-1} + \eta_M e_{t-1}^{\xi_e}. \tag{32}$$

The productive-capability stock $\overline{A}_t$ accumulates from the building-effort flow x_t with efficiency η_A, curvature $\xi_x \in (0,1)$, and depreciation δ_A. The term $\mathcal{S}_t^A$ allows market size in the strategic sector to raise the effectiveness of implementation; the benchmark sets the elasticity $\phi_{AX} = 0.05$, thus the accelerator is modest and equals one in steady state. Customer capital accumulates from lagged market-building with efficiency η_M and curvature $\xi_e \in (0,1)$ and depreciates at the policy-controlled rate $\delta_{m,t}$. The one-period customer-capital timing separates immediate customer acquisition in demand from the durable market-power state. Organizational capital depreciates at rate δ_H and is rebuilt through investment i_t^H.

The general specification also allows separable convex costs for market-building, organizational investment, and productive capability:

$$C_t^E = \frac{\Psi_E}{1 + \xi_e} e_t^{1+\xi_e}, \tag{33}$$

$$C_t^H = \frac{\Psi_H}{1 + \xi_H} (i_t^H)^{1+\xi_H}, \tag{34}$$

$$C_t^X = \frac{\Psi_X}{1 + \xi_x} x_t^{1+\xi_x}. \tag{35}$$

The benchmark calibration uses $\Psi_H = 0.0075$, $\Psi_X = 0$, and sets the market-building flow-cost parameter to zero; the smooth throughput cost Γ_t supplies the main organizational discipline.

Market-building outlays are also exposed to a financing wedge. Let q_t^L denote the price

of a long-duration claim used to finance customer-acquisition outlays:

$$q_t^L = \mathbb{E}_t\left[\text{disc}_{t,t+1}\{1+\kappa_L q_{t+1}^L\}\right], \tag{36}$$

$$s_t^e = \rho_e^s s_{t-1}^e + \chi_R^s \varepsilon_t^R, \tag{37}$$

$$\mathcal{F}_t^e = \left(\frac{Q_L^{ss}}{q_t^L}\right)^{\phi_{fin,e}} \exp\left(\phi_{fin,e} D_L s_t^e/400\right), \tag{38}$$

$$\mathcal{F}_t^x = \left(\frac{Q_L^{ss}}{q_t^L}\right)^{\phi_{fin,x}} \exp\left(\phi_{fin,x} D_L s_t^e/400\right). \tag{39}$$

Here κ_L is the coupon parameter that gives the claim duration D_L, Q_L^{ss} is its steady-state price, s_t^e is the customer-acquisition financing spread, and (ρ_e^s, χ_R^s) govern its persistence and its loading on monetary-policy innovations. The benchmark sets $\phi_{fin,x} = 0$ and $\phi_{fin,e} = 0.15$. Thus monetary easing affects market-building not only through demand and discounting, but also by lowering the effective financing cost of customer acquisition; the productive-capability finance exposure is inactive in the benchmark.

The F.O.Cs are

$$\eta_A \xi_x x_t^{\xi_x - 1} \mathcal{S}_t^A q_t^P = \mathcal{F}_t^x \nu_t^H + \Psi_X x_t^{\xi_x}, \tag{40}$$

$$\eta_M \xi_e e_t^{\xi_e - 1} \mathbb{E}_t[\text{disc}_{t,t+1}\, q_{t+1}^m] + \zeta_e p_{S,t} Y_{S,t} \frac{\phi_m}{d_t \varepsilon_t^2} = \omega_e \nu_t^H + \mathcal{F}_t^e \kappa_e + \tau_{r,t} + \Psi_E e_t^{\xi_e}, \tag{41}$$

$$1 + \Psi_H (i_t^H)^{\xi_H} = \mathbb{E}_t\Bigg[\text{disc}_{t,t+1}\left(\psi_O \frac{\phi_O}{1+\xi_O} T_{t+1}^{1+\xi_O} H_t^{-\psi_O - 1} + (1-\delta_H)\{1+\Psi_H (i_{t+1}^H)^{\xi_H}\}\right)\Bigg]. \tag{42}$$

Condition (40) equates the marginal value of productive-capability implementation to the effective marginal cost of the throughput it uses. The marginal value is

$$\eta_A \xi_x x_t^{\xi_x - 1} \mathcal{S}_t^A q_t^P.$$

Condition (41) equates the value of market-building, its discounted contribution to future customer capital q_{t+1}^m plus its contemporaneous demand effect, to the marginal throughput cost scaled by the organizational intensity ω_e, the financing-adjusted operating cost $\mathcal{F}_t^e \kappa_e$, and the competition tax $\tau_{r,t}$. The H Euler equation prices organizational investment by the future reduction in convex throughput costs. This is the mechanism that makes organizational investment rise with organizational capital after a demand expansion rather than move mechanically in the opposite direction.

3.5 Shadow values

The shadow values of productive capability and customer capital are

$$q_t^A = p_{S,t} Y_{S,t} \frac{\mu_t^* - 1}{\mu_t^*} \frac{1}{\overline{A}_t} + \mathbb{E}_t \left[\text{disc}_{t,t+1} (1 - \delta_A) q_{t+1}^A \right], \tag{43}$$

$$q_t^B = \mathbb{E}_t [\text{disc}_{t,t+1} \, q_{t+1}^A], \tag{44}$$

$$q_t^P = q_t^A, \tag{45}$$

$$q_t^m = p_{S,t} Y_{S,t} \left(\Gamma_M + \Lambda_M z_t^B + \Lambda_{ZM} z_t^M + \frac{\phi_m}{d_t \varepsilon_t^2} + v_M gap_t \right) + \mathbb{E}_t [\text{disc}_{t,t+1} (1 - \delta_{m,t}) q_{t+1}^m]. \tag{46}$$

Equation (43) values productive capability at its current marginal revenue product, markup-adjusted revenue $p_{S,t} Y_{S,t} (\mu_t^* - 1)/\mu_t^*$ per unit of $\overline{A}_t$, plus its depreciated continuation value. Since current $\overline{A}_t$ enters current production, current productive investment is priced by $q_t^P = q_t^A$; q_t^B is retained only as the one-period delayed value object used in timing comparisons. In the customer-capital value (46), Γ_M is the baseline per-period rent from a marketed customer, $\phi_m/(d_t \varepsilon_t^2)$ the marginal markup rent, and $v_M \geq 0$ loads the output gap gap_t so that customers are more valuable in booms. In the benchmark calibration, Λ_M and Λ_{ZM} are set to zero, thus demand affects the customer-capital block mainly through final-demand composition, current market-building in d_t, and the output-gap loading. Positive Λ_{ZM} is a private-rent channel: it raises the firm's payoff to customer capital when the market-power state is high, without turning that state into a planner technology shock.

3.6 Capital, policy, and feasibility

Sectoral capital evolves according to

$$K_{S,t} = (1 - \delta_K) K_{S,t-1} + i_{S,t}^K, \tag{47}$$

$$K_{O,t} = (1 - \delta_K) K_{O,t-1} + i_{O,t}^K. \tag{48}$$

The corresponding Tobin's-q equations use sector-specific adjustment costs:

$$Q_{S,t}^K = 1 + \phi_{K,S} \left(i_{S,t}^K / K_{S,t-1} - \delta_K \right), \tag{49}$$

$$Q_{O,t}^K = 1 + \phi_{K,O} \left(i_{O,t}^K / K_{O,t-1} - \delta_K \right). \tag{50}$$

Capital in each sector depreciates at rate δ_K and faces a quadratic investment-adjustment cost with sector-specific weights $\phi_{K,S}, \phi_{K,O} \geq 0$; $Q_{j,t}^K$ is the corresponding Tobin's q, and

$i^K_{j,t}$ is sectoral investment. Appendix B records the full continuation-value equations.

Monetary policy follows

$$\log \frac{R_t}{R^{ss}} = \rho_R \log \frac{R_{t-1}}{R^{ss}} + (1-\rho_R)\Bigg[\phi_\pi \log \frac{\pi_t}{\pi^{ss}} + \phi_y(\log Y_t - \log Y^{ss}) + \phi_{\Delta y}(\log Y_t - \log Y_{t-1})\Bigg] + \varepsilon^R_t. \tag{51}$$

where ρ_R is the degree of interest-rate smoothing, ϕ_π the response to inflation, ϕ_y and $\phi_{\Delta y}$ the responses to the output level and to output growth (a speed-limit term), and ε^R_t is a monetary-policy shock. The rule responds to the observed output cycle $\log Y_t - \log Y^{ss}$ and to output growth. This object differs from the natural gap used in the customer-capital valuation and competition-policy rule,

$$gap_t = \log Y_t - \log Y^{nat}_t. \tag{52}$$

The latter measures the cyclical value of market-building relative to the flexible-price allocation. The natural allocation is the flexible-price version of the same two-sector economy. The natural block keeps the same multiplicative z^M_t elasticity schedule but omits the private-rent loading $\Lambda_{ZM} z^M_t$ and the output-gap loading $v_M gap_t$ in the customer-capital value equation, since the output gap is an actual-economy object that is zero in the natural allocation by construction.

Competition policy has two instruments. The first is the depreciation rate of customer capital, $\delta_{m,t}$: a higher rate erodes incumbents' inherited demand power more quickly. The second is a proportional tax $\tau_{r,t}$ on market-building effort, which raises its private marginal cost in condition (41). Both follow inertial feedback rules that lean against deviations of customer capital and market-building from their natural-allocation counterparts and against the output gap:

$$\delta_{m,t} = (1-\rho_\delta)\delta^{ss}_m + \rho_\delta \delta_{m,t-1} + \phi_{\delta,m}(\bar{m}_t - \bar{m}^{nat}_t) + \varepsilon^\delta_t, \tag{53}$$

$$\tau_{r,t} = \tau^{ss}_r + \phi_{\tau,e}(e_t - e^{nat}_t) + \phi_{\tau,y} gap_t + \phi_{\tau,z} z^M_t + \varepsilon^\tau_t. \tag{54}$$

Here ρ_δ is the persistence of the depreciation rule, $\phi_{\delta,m}$ and $(\phi_{\tau,e}, \phi_{\tau,y}, \phi_{\tau,z})$ are feedback coefficients, $\bar{m}^{nat}_t$ and e^{nat}_t are natural-allocation counterparts, and $\varepsilon^\delta_t, \varepsilon^\tau_t$ are policy innovations. In the benchmark these rules are inactive (all feedback coefficients are zero); Section 8 optimizes them. Regulation uses resources:

$$G^{reg}_t = \frac{\omega_\delta}{2}(\delta_{m,t} - \delta^{ss}_m)^2 + \frac{\omega_\tau}{2}(\tau_{r,t} - \tau^{ss}_r)^2 + \frac{\bar{\omega}_\delta}{2}(\delta^{ss}_m - \delta^{ref}_m)^2 + \frac{\bar{\omega}_\tau}{2}(\tau^{ss}_r - \tau^{ref}_r)^2. \tag{55}$$

The first two terms penalize cyclical use of the instruments (weights $\omega_\delta, \omega_\tau$); the last

two penalize permanent departures of the long-run stance from institutional reference levels $\delta_m^{ref}, \tau_r^{ref}$ (weights $\bar{\omega}_\delta, \bar{\omega}_\tau$), so the benchmark long-run stance is held fixed. The five structural disturbances—TFP z_t^A, demand z_t^B, market power z_t^M, wage markup z_t^W, and price markup z_t^P—each follow an AR(1) process; their innovations, together with the monetary shock ε_t^R, are the model's stochastic drivers. The aggregate resource constraint is

$$
\begin{aligned}
Y_t = {} & C_t + i_{S,t}^K + i_{O,t}^K + i_t^H + \Gamma_t + C_t^H + C_t^X + C_t^E \\
& + \frac{\phi_{K,S}}{2}\left(\frac{i_{S,t}^K}{K_{S,t-1}} - \delta_K\right)^2 K_{S,t-1} + \frac{\phi_{K,O}}{2}\left(\frac{i_{O,t}^K}{K_{O,t-1}} - \delta_K\right)^2 K_{O,t-1} \\
& + \frac{\theta}{2}(\pi_{S,t} - 1)^2 p_{S,t} Y_{S,t} && (56) \\
& + \frac{\theta_w}{2}(\pi_{w,t} - 1)^2 w_t N_t && (57) \\
& + G_t^{reg}. && (58)
\end{aligned}
$$

4 Equilibrium and Analytical Results

Definition 1 (Decentralized equilibrium)**.** A decentralized equilibrium is a collection of processes for household allocations, sectoral quantities and prices, intangible stocks and controls, shadow values, policy instruments, and natural-allocation counterparts such that households satisfy Equations (2)–(5); firms satisfy Equations (8)–(46); sectoral capital, customer capital, productive capability, and organizational capital follow their laws of motion; policy follows Equations (51)–(54); and feasibility (58) holds.

Define

$$
\begin{aligned}
\widetilde{q}_t^m &\equiv \mathbb{E}_t[\mathrm{disc}_{t,t+1}\, q_{t+1}^m], && (59) \\
\mathcal{Q}_t^A &\equiv \mathcal{S}_t^A q_t^P, && (60) \\
\mathcal{N}_t^x &\equiv \mathcal{F}_t^x \nu_t^H, \qquad \mathcal{C}_t^e \equiv \mathcal{F}_t^e \kappa_e, && (61) \\
\mathcal{D}_t &\equiv \zeta_e p_{S,t} Y_{S,t} \frac{\phi_m}{d_t \varepsilon_t^2}. && (62)
\end{aligned}
$$

The local intangible block is

$$
\begin{aligned}
\eta_A \xi_x x_t^{\xi_x - 1} \mathcal{Q}_t^A &= \mathcal{N}_t^x + \Psi_X x_t^{\xi_x}, && (63) \\
\eta_M \xi_e e_t^{\xi_e - 1} \widetilde{q}_t^m + \mathcal{D}_t &= \omega_e \nu_t^H + \mathcal{C}_t^e + \tau_{r,t} + \Psi_E e_t^{\xi_e}. && (64)
\end{aligned}
$$

Proposition 1 (Local monotonicity of the smooth-throughput intangible block)**.** *Sup-*

pose $0 < \xi_x < 1$, $0 < \xi_e < 1$, $\Psi_X \geq 0$, $\omega_e > 0$, *and*

$$\frac{\partial}{\partial e_t}\left[\eta_M \xi_e e_t^{\xi_e - 1} \widetilde{q}_t^m + \mathcal{D}_t - \Psi_E e_t^{\xi_e}\right] < 0 \tag{65}$$

locally. Then productive capability building is locally increasing in $\mathcal{Q}_t^A$ *and decreasing in* $\mathcal{N}_t^x$ *and* Ψ_X*. Market-building is locally increasing in* $\widetilde{q}_t^m$ *and* $\mathcal{D}_t$*, and decreasing in* v_t^H*,* $\tau_{r,t}$*,* ω_e*, and the financing wedge* $\mathcal{F}_t^e$ *when* $\kappa_e > 0$*.*

Proof. See Appendix C. □

Corollary 1 (Removal of the hard-capacity restriction)**.** *In the smooth-throughput economy, predetermined* H_{t-1} *affects the marginal cost* v_t^H *but does not impose* $dx_t + \omega_e de_t = 0$ *on impact. Hence the model does not mechanically force productive capability building to fall whenever market-building rises. A hard-capacity formulation is recovered only as a limiting case in which the throughput cost becomes an infinite penalty away from* $T_t = H_{t-1}$*.*

Proof. See Appendix C. □

Proposition 2 (Competition policy lowers customer-capital building)**.** *In a neighborhood where Proposition 1 holds, a higher market-building tax* $\tau_{r,t}$ *lowers* e_t*. A higher expected customer-capital depreciation rate lowers the continuation value of customer capital and therefore lowers* e_t*. If this reduction relaxes the organizational capacity shadow value, productive capability building* x_t *rises.*

Proof. See Appendix C. □

Proposition 3 (Demand accommodation and intangible reallocation)**.** *Let a disturbance raise the customer-building composite*

$$\mathcal{M}_t = \eta_M \xi_e e_t^{\xi_e - 1} \widetilde{q}_t^m + \mathcal{D}_t$$

relative to the productive-building composite

$$\mathcal{P}_t = \eta_A \xi_x x_t^{\xi_x - 1} \mathcal{Q}_t^A - \Psi_X x_t^{\xi_x}.$$

If the induced movement in v_t^H *is not large enough to offset the increase in either composite, the corresponding intangible margin rises. In the benchmark calibration, a demand expansion raises both* $\mathcal{M}_t$ *and* $\mathcal{P}_t$ *on impact: productive-capability building and market-building both rise, with the customer-capital stock responding more persistently. A monetary easing also raises both intangible margins on impact, but its customer-capital effect is smaller and less persistent because it works indirectly through the policy-rate, demand, and inflation blocks. The model object is the innovation* ε_t^R *in the Taylor rule (51). Section 6 brings this object to*

the data using high-frequency FOMC event-study surprises, aggregated to the firm fiscal year and signed so that larger values denote unexpected monetary accommodation. The demand specification separately isolates the expansionary force that carries the persistent customer-capital response.

Proof. See Appendix C. □

Proposition 4 (Planner wedge against customer-capital accumulation)**.** *Relative to the decentralized condition* (64)*, the cooperative planner assigns an additional shadow cost* λ_t^{ext} *to market-building:*

$$\eta_M \xi_e e_t^{\xi_e - 1} \widetilde{q}_t^m + \mathcal{D}_t = \omega_e \nu_t^H + \kappa_e + \tau_{r,t} + \Psi_E e_t^{\xi_e} + \lambda_t^{ext}. \tag{66}$$

If $\phi_m > 0$*,* $\varepsilon_t > 1$*, and the strategic-sector markup wedge is positive, then the allocative component of* λ_t^{ext} *is strictly positive. Price-adjustment losses and inefficient organizational reallocation add nonnegative terms when they are locally increasing in customer-capital accumulation.*

Proof. See Appendix C. □

Corollary 2 (Local complementarity of monetary and competition policy)**.** *When competition policy lowers* $\mathcal{M}_t$*, the marginal need for monetary policy to lean against market-power accumulation is smaller. Conversely, stronger monetary stabilization lowers the marginal value of cyclical competition intervention. The instruments are therefore local complements around the cooperative allocation.*

Proof. See Appendix C. □

The analytical model isolates the customer-capital margin evaluated below. The benchmark calibration satisfies the key real-side sign restrictions: demand expansions raise output and organizational investment, and the market-power state remains active. The main quantitative tradeoff is the joint mapping from aggregate demand into output amplification and measured labor-share volatility.

5 Quantitative Discipline

The benchmark is a quarterly U.S. calibration. Standard preference, capital, and monetary-policy parameters are chosen from the New Keynesian literature. The new customer-capital and organizational-throughput parameters are disciplined by firm-level measures of revenue shares, productive intangible expenditures, and broad operating-cost

proxies for market-building, by steady-state labor-share and markup targets, and by business-cycle moments for output, inflation, policy rates, labor shares, productivity growth, and markups. Appendix A reports the construction of firm-level model objects and the full calibration.

The discipline has two layers. First, firm-level accounting data pin down the steady-state scale of the strategic sector and the relative size of productive and market-building-related operating expenditures. Second, the stochastic block is chosen to match a small set of aggregate business-cycle moments, always using the same cyclical transformation in the data and in simulated series. This separation matters because the paper's central result is not obtained from a large variance contribution of an unobserved markup shock; it comes from the endogenous response of firms' market-building decisions to demand and monetary conditions.

Table 1 reports the business-cycle moment fit. The labor-share target is the HP-filtered cyclical labor share, measured in the same way in the data and the simulated model. This common cyclical transformation is important because measured labor shares contain low-frequency movements that are outside the stationary business-cycle model.

Table 1: Business-cycle moment fit

Moment / shock	U.S. target	Model
Std. GDP cycle	1.195	1.077
Std. inflation	0.975	1.003
Std. fed funds	1.083	1.057
Std. labor share cycle	0.948	1.119
Std. annual TFP growth	0.992	0.803
Markup/output volatility ratio	0.400	0.505
Markup variance share orth. to y, π (lit.)	0.500	0.528
Corr. GDP cycle, inflation	0.341	0.403
Corr. GDP cycle, fed funds	0.726	0.768
σ_R	–	0.0015
σ_A	–	0.0050
σ_B	–	0.0421
σ_M	–	0.0150
σ_W	–	0.0200
σ_P	–	0.0062

6 Firm-Level Evidence

Although the paper's main quantitative results come from the structural model, the mechanism has a sharp cross-sectional implication. If customer-capital-intensive firms

have a higher payoff to acquiring customers in expansions, then aggregate conditions should shift their intangible investment different from other similar firms. We evaluate this implication in a firm-year panel of U.S. Compustat firms (1999–2025, excluding financials and utilities) with direct measures of intangible expenditure. The exposure sample contains 7,134 firms and 74,029 firm-years before outcome- and macro-series-specific missingness. Market-building is selling, general, and administrative expenditure net of R&D, scaled by sales: the accounting measure of resources a firm devotes to selling, marketing, distribution, and customer acquisition. This measure follows directly from work that treats SG&A as a productive input rather than as overhead. Traina (2018) shows that SG&A consists mainly of marketing and management, "indirect inputs to production" whose share of firm costs have risen secularly, and that accounting for it materially changes measured markups; Eisfeldt and Papanikolaou (2013) capitalize SG&A spending into a stock of organisational capital. We accordingly read SG&A net of R&D as investment in customer capital, the demand-side analogue of the technology investment in R&D, and thus as precisely the margin that the production-approach markup literature treats as overhead even as measured market power rises (De Loecker et al., 2020; Karabarbounis and Neiman, 2019; Corrado et al., 2009). Productive capability is R&D, the empirical counterpart of the model's capability-building margin x; physical capital expenditure, which maps to the model's separate physical-investment block, is examined separately.

We use an exposure design that interacts aggregate shocks with a predetermined measure of the firm's market-building intensity. Let

$$e_{it}^{data} = 100 \times \frac{\text{SG\&A}_{it} - \text{R\&D}_{it}}{\text{Sales}_{it}}$$

denote the firm-year market-building ratio after the sample filters and accounting-ratio treatment described in Appendix A. For each firm, we average this ratio over the first five sample years in which exposure is assigned,

$$\tilde{e}_i = \frac{1}{|T_i^0|} \sum_{t \in T_i^0} e_{it}^{data}, \qquad T_i^0 = \{1999, \ldots, 2003\} \cap T_i,$$

and standardise the resulting cross-sectional measure,

$$\bar{e}_i = \frac{\tilde{e}_i - \mathbb{E}_j[\tilde{e}_j]}{\text{sd}_j(\tilde{e}_j)}.$$

Thus $\bar{e}_i$ is fixed before the local-projection response window and is measured in cross-

sectional standard-deviation units. We then estimate the dynamic specification

$$\begin{aligned} y_{i,t+h} - y_{i,t-1} = \beta_{D,h}\,(g_t \times \bar{e}_i) + \beta_{R,h}\,(s_t \times \bar{e}_i) \\ + \mathbf{X}'_{i,t-1}\Gamma_h + \alpha_i^h + \lambda_t^h + u_{i,t+h}, \end{aligned} \tag{67}$$

for horizons $h = 0, \ldots, 5$, with both aggregate interactions in the same dynamic equation. The outcome $y_{i,t+h} - y_{i,t-1}$ is the cumulative change in one of four firm-level objects. The two primary outcomes are log market-building expenditure and log R&D expenditure, the direct empirical counterparts of the model's market-building and productive-capability margins. Two auxiliary outcomes discipline the economic interpretation: log physical capital expenditure checks whether the response is specific to intangible margins rather than physical investment, and EBITDA/sales measures operating profitability. The main table reports the two primary margins; the figures (Figures 3–4) add physical capital expenditure and the profitability diagnostic. The aggregate-demand variable is $g_t = \{100\Delta \log(GDP_t) - \bar{g}\}/\sigma_g$, standardized annual real GDP growth. It measures the economy-wide expansion in expenditure and sales opportunities faced by firms, and is the empirical analogue of the model's demand expansion. The monetary variable s_t is a standardised monetary-easing innovation, discussed in the next paragraph. The terms α_i and λ_t are firm and year fixed effects. The control vector $\mathbf{X}_{i,t-1}$ contains lagged log assets, lagged leverage, and lagged cash/assets. Because year effects absorb the aggregate shocks themselves, $\beta_{D,h}$ and $\beta_{R,h}$ are identified from the differential response of firms with different pre-period exposure. To ensure that this differential reflects the customer-capital channel rather than cyclical heterogeneity in size, financing, or sector, the baseline specification additionally: i) interacts each aggregate shock with the firm controls and; ii) replaces year fixed effects with industry$\times$year fixed effects, so that identification is within industry and net of shock-by-characteristic heterogeneity.

We measure s_t using high-frequency changes in financial-market prices around FOMC communications. Specifically, we use the combined monetary-event surprise from the FOMC-communication database of Acosta et al. (2025), who measure asset-price changes in narrow windows around FOMC announcements and distinguish the policy statement and press-conference components. The series is aggregated to the firm fiscal year and signed so that larger values denote unexpected monetary accommodation. Because the measure is based on event-window asset-price changes rather than realised changes in the policy rate, it remains informative during zero lower bound episodes. We assess robustness to a rule-based definition of monetary innovations using a Taylor-rule residual: with Δi_t the annual change in the federal funds rate, g_t real GDP growth, and π_t^c core inflation, we estimate

$$\Delta i_t = a_R + \rho_i \Delta i_{t-1} + \gamma_0 g_t + \gamma_1 g_{t-1} + \delta_0 \pi_t^c + \delta_1 \pi_{t-1}^c + u_t^R, \tag{68}$$

and set the standardized rule-based easing innovation to

$$s_t^{\text{rule}} = -\hat{u}_t^R / \text{sd}(\hat{u}_t^R). \tag{69}$$

The two measures are positively correlated ($\rho \approx 0.41$) and deliver same-signed coefficients in every specification (Panel B of Tables 2 and 3). We use the high-frequency surprise as the main measure because it remains a clean innovation at the zero lower bound, whereas the rule residual is mechanically tied to realized policy and so comoves with the demand cycle; as we discuss below, this inflates the apparent monetary effect under the rule-based measure. All ratio outcomes are trimmed to remove near-zero-sales observations and extreme accounting ratios.

The regressor of interest is an aggregate shock interacted with cross-sectional exposure, thus its identifying variation comes from the aggregate time series, about twenty-six years, rather than from the cross-section alone. We therefore report inference under four approaches: standard errors clustered by firm, standard errors clustered two-way by firm and year, the wild-cluster bootstrap on the year dimension (Cameron et al., 2008), and a leave-one-year-out cluster jackknife. The Wild bootstrap and the jackknife are the most reliable when the number of year clusters is small, and a few aggregate episodes dominate the shock series, and we treat them as our primary inference; the firm-clustered and two-way estimates are reported alongside for comparison across methods.

Table 2 focuses on the two model-mapped intangible-investment margins, market building and productive-capability building (R&D), and reports their local-projection responses to aggregate demand and to high-frequency monetary easing, at horizons $h = 1$ and $h = 2$, under industry$\times$year fixed effects and under both control configurations (aggregate shocks interacted with firm controls, and level controls only). For every coefficient, we report the two-sided p-value under all four inference methods.

One response is robust to every conservative inference method and to both control configurations: aggregate demand expansions raise market-building among customer-capital-intensive firms. A one-standard-deviation demand expansion interacted with a one-standard-deviation exposure difference raises market building by 1.2 to 1.4 log points at the two-year horizon, with two-sided p-values at or below 0.03 under the wild-cluster bootstrap and below 0.001 under the leave-one-year-out jackknife. The productive-capability response is positive but less precisely estimated, reaching significance under the jackknife in the two-year specification but not under the wild-cluster bootstrap. This is the dynamic cross-sectional counterpart of the model's central mechanism in Figure 8, and it is the firm-level moment we use to discipline the model's customer-capital channel: firms organized around market-building tilt toward customer acquisition precisely when aggregate demand raises its return. Here, the four

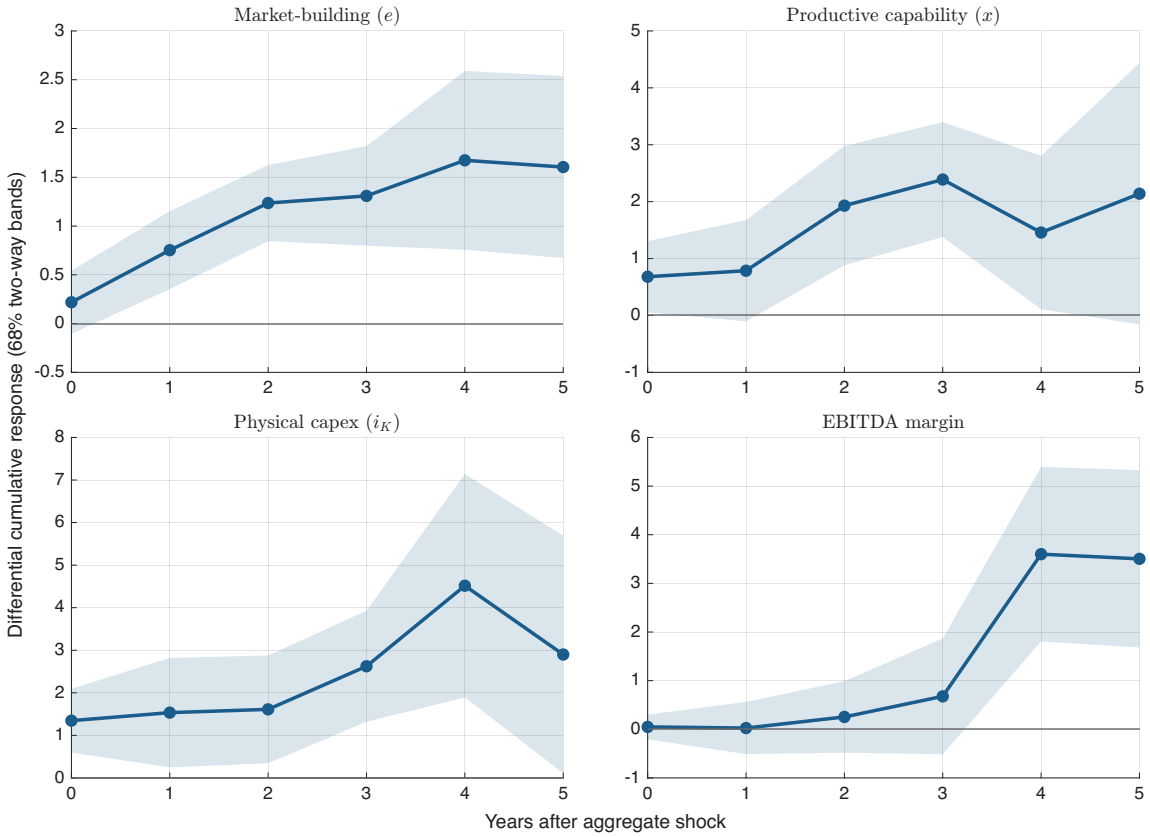
Table 2: Firm-level local projections: demand and monetary responses of intangible investment

					p-value by inference method			
Outcome	Controls	h	Term	$\hat{\beta}$	firm-cl.	two-way	wild boot.	jackknife
Panel A. High-frequency monetary surprise (USMPD)								
Market-building (log e)	$\times$ shocks	1	Demand	0.75	0.057*	0.060*	0.077*	0.168
		1	Monetary	1.02	0.111	0.061*	0.104	0.105
		2	Demand	1.24	0.017**	0.002***	0.027**	<.001***
		2	Monetary	0.57	0.474	0.392	0.393	0.411
	levels only	1	Demand	1.05	0.005***	0.010***	0.026**	0.057*
		1	Monetary	0.98	0.099*	0.072*	0.082*	0.115
		2	Demand	1.40	0.004***	<.001***	0.043**	<.001***
		2	Monetary	0.27	0.727	0.662	0.538	0.604
Productive capability (log R&D)	$\times$ shocks	1	Demand	0.79	0.393	0.377	0.312	0.315
		1	Monetary	0.35	0.809	0.810	0.793	0.806
		2	Demand	1.93	0.101	0.065*	0.124	0.058*
		2	Monetary	1.66	0.391	0.356	0.320	0.306
	levels only	1	Demand	0.67	0.378	0.437	0.472	0.469
		1	Monetary	0.44	0.715	0.754	0.753	0.763
		2	Demand	1.98	0.038**	0.032**	0.172	0.040**
		2	Monetary	2.15	0.194	0.190	0.198	0.177
Panel B. Taylor-rule residual (rule-based monetary measure)								
Market-building (log e)	$\times$ shocks	1	Demand	0.38	0.274	0.223	0.154	0.204
		1	Monetary	0.60	0.127	0.142	0.212	0.224
		2	Demand	1.06	0.011**	0.003***	0.012**	0.001***
		2	Monetary	0.68	0.058*	0.003***	0.074*	0.121
	levels only	1	Demand	0.75	0.019**	0.026**	0.051*	0.047**
		1	Monetary	0.95	0.007***	0.024**	0.043**	0.068*
		2	Demand	1.40	<.001***	<.001***	0.030**	<.001***
		2	Monetary	1.07	<.001***	0.005***	0.009***	0.065*
Productive capability (log R&D)	$\times$ shocks	1	Demand	0.67	0.326	0.206	0.206	0.179
		1	Monetary	0.27	0.744	0.660	0.638	0.670
		2	Demand	1.25	0.133	0.085*	0.062*	0.091*
		2	Monetary	0.77	0.384	0.124	0.308	0.395
	levels only	1	Demand	0.69	0.286	0.208	0.204	0.182
		1	Monetary	1.37	0.059*	0.018**	0.032**	0.033**
		2	Demand	1.21	0.105	0.046**	0.040**	0.026**
		2	Monetary	1.76	0.023**	<.001***	0.021**	0.026**

Notes. Each row is the coefficient on the indicated aggregate shock interacted with pre-period market-building exposure, from the joint local projection (67) at horizon h, with firm and industry$\times$year fixed effects. Panel A uses the high-frequency USMPD monetary surprise (the baseline measure); Panel B replaces it with the Taylor-rule residual (69). The demand coefficients are essentially unchanged across the two measures; the monetary coefficients are not. Coefficients $\hat{\beta}$ are per standard-deviation shock $\times$ standard-deviation exposure, in percent (100 $\times$ log dollars). "$\times$ shocks" interacts the firm controls (lagged log assets, leverage, cash/assets) with each aggregate shock; "levels only" enters them in levels. The four columns report the two-sided p-value for the same coefficient under firm-clustered standard errors, two-way (firm and year) clustered standard errors, the wild-cluster bootstrap on year ($B = 1999$), and the leave-one-year-out cluster jackknife. Because the identifying variation is at the year level, we treat the wild bootstrap and jackknife as our primary inference; the firm-clustered and two-way p-values are reported alongside for comparison. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

inference methods agree, the demand response is significant even under firm clustering, the variance estimator therefore does not drive the conclusion.

Figure 3: Firm-level responses to demand expansions

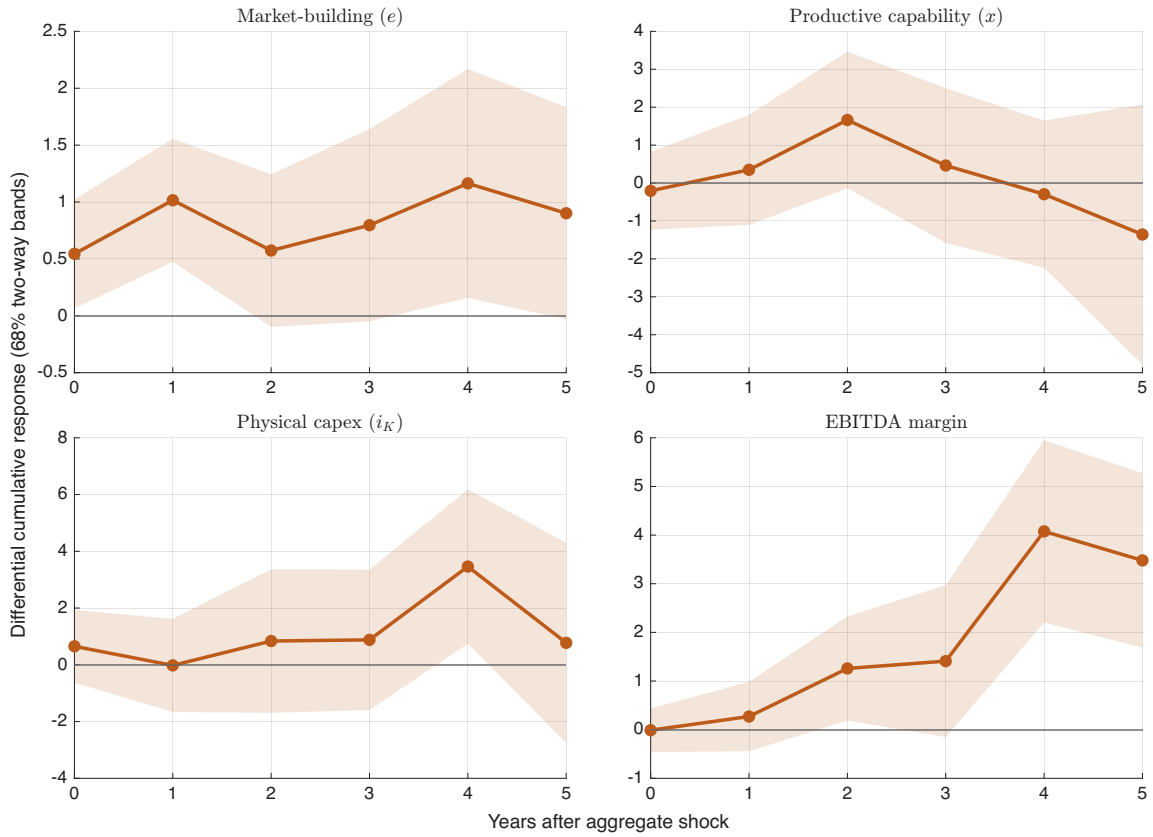


Note: The figure plots $\beta_{D,h}$ from the joint local projection (67) under industry×year fixed effects, for the three investment margins, market building (e), productive capability (R&D), and physical capital expenditure (i_K), together with the EBITDA margin. Shaded bands are 68 percent two-way clustered confidence bands.

The monetary differential is weaker and sensitive to how the shock is measured. Under the clean high-frequency surprise (Panel A) the monetary-easing coefficients are positive and correctly signed, but not robustly identified: the identifying variation is concentrated in a few episodes (2001 and 2008), and the wild-bootstrap and jackknife p-values stay above conventional levels. Under the Taylor-rule residual (Panel B) the same positive sign appears throughout (Figure 5), and for market-building, it becomes significant: the two-year response reaches the one percent level in the levels-only specification. We read this asymmetry not as a stronger monetary result, but as a measurement artifact: the rule residual co-moves with the demand cycle, thus it absorbs part of the robust demand channel into the monetary coefficient, whereas the high-frequency surprise is orthogonal to the cycle by construction. We therefore treat the firm-level monetary evidence as sign corroboration of the model's monetary channel rather than a clean causal estimate: the structural monetary channel is identified from the aggregate moments and the policy block, not from the monetary cross-section, and a sharper

firm-level test would require a within-year exposure design, which we leave to future work.

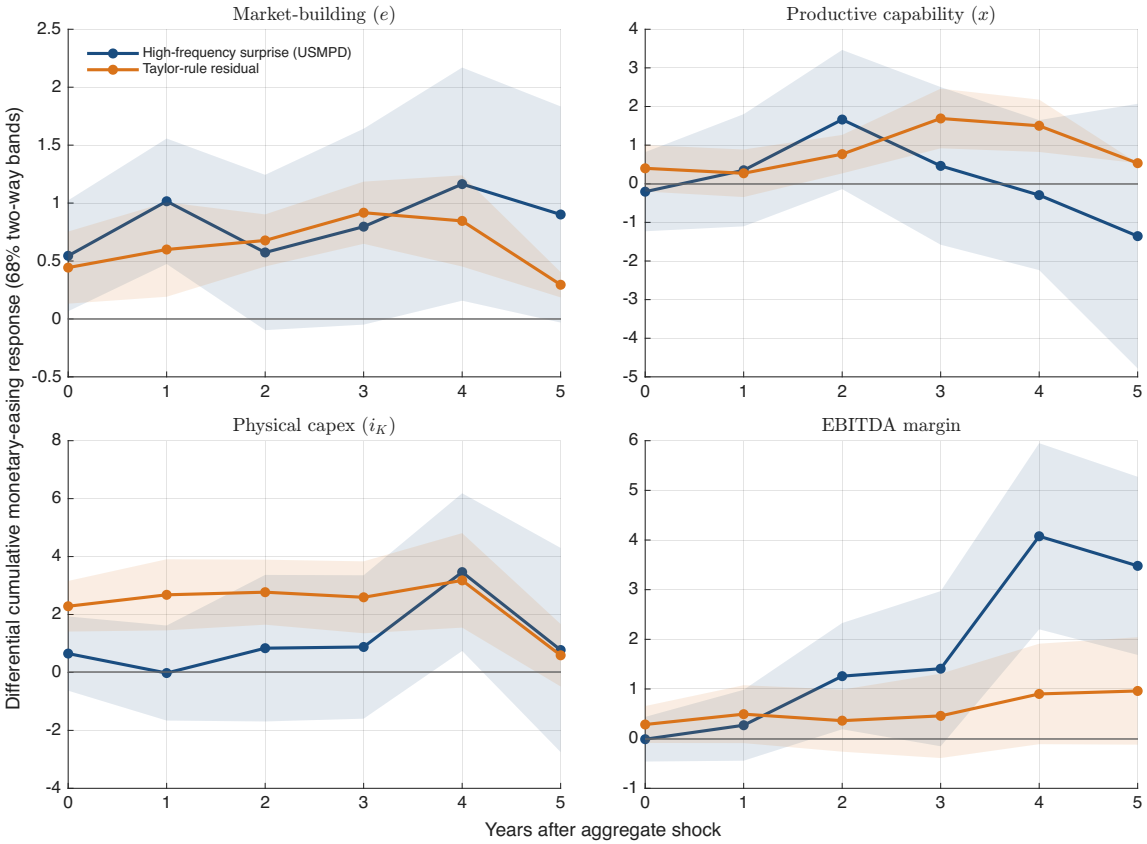
Figure 4: Firm-level responses to high-frequency monetary easing



Note: The figure plots $\beta_{R,h}$ from the joint local projection (67) for the three investment margins—market-building (e), productive capability (R&D), and physical capital expenditure (i_K)—together with the EBITDA margin. Shaded bands are 68 percent two-way clustered confidence bands; the wide bands reflect the limited number of identifying monetary episodes, and the monetary differential is not robustly identified at annual frequency.

Because exposure is a pre-period firm average, one concern is that the demand result reflects the timing of exposure rather than a predetermined firm characteristic. The baseline already begins the shock window in 2004, after the 1999–2003 exposure window; we widen the separation further by restricting the shocks to begin in 2007 or 2008, so that several years pass between the measurement of exposure and the response window. Table 3 reports the headline $h = 2$ coefficients under all four inference methods, and Figures 6–7 plot the corresponding response paths. The demand response of market-building is essentially unchanged: the coefficient stays near 1.2 and remains significant under the wild-cluster bootstrap and the jackknife as the gap widens—weakening only mildly at the 2008 start, where fewer identifying years remain——and its response path stays within the baseline confidence band. The monetary channel shows the same measure-dependence as in Table 2: weak under the high-frequency surprise and marginally significant under the Taylor residual, across

Figure 5: Firm responses to monetary easing under two monetary measures



Note: The figure plots $\beta_{R,h}$ for the four firm-level outcomes under the high-frequency USMPD surprise and under the Taylor-rule residual. Shaded regions are 68 percent two-way clustered confidence bands. The market-building response carries the same positive sign under both measures; the rule-based measure, which co-moves with the demand cycle, attributes a larger response to monetary easing.

the timing variants. The demand-to-market-building result is therefore not an artifact of exposure timing.

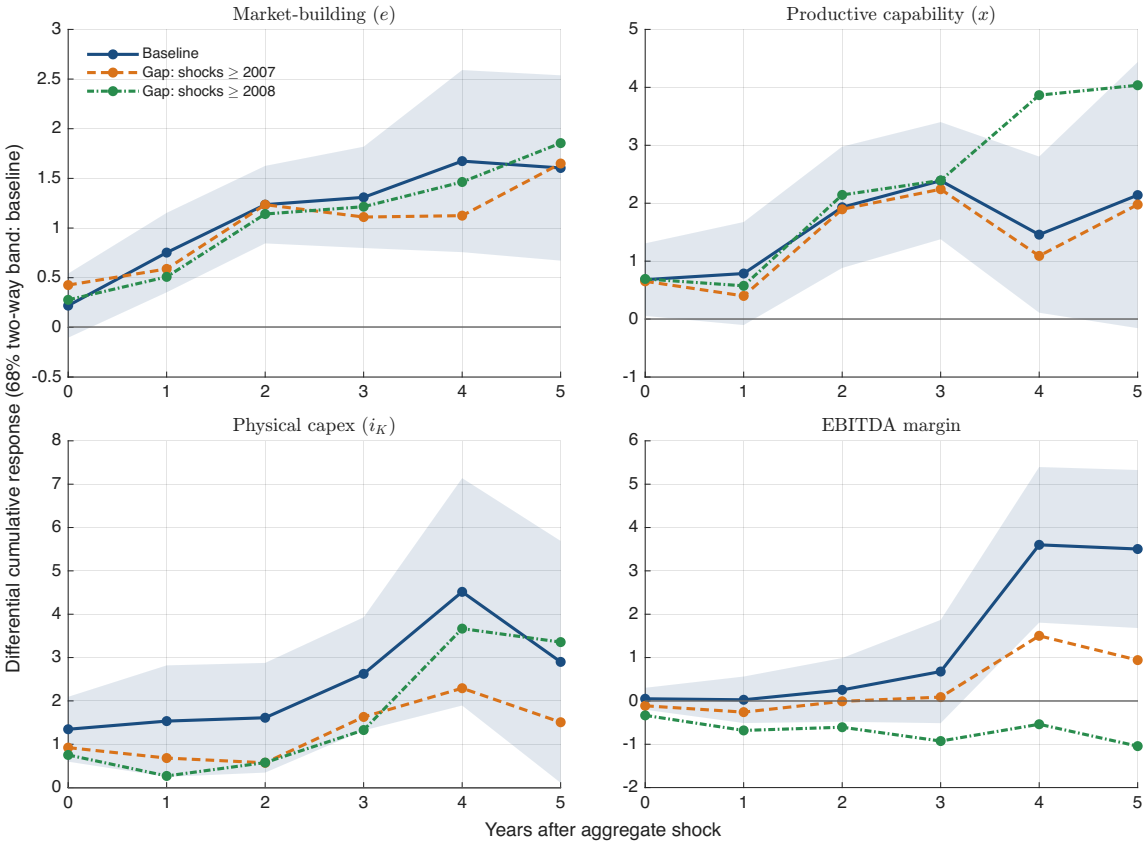
Table 3: Exposure-timing robustness of the firm-level local projections

				p-value by inference method			
Specification	Outcome	Term	$\hat{\beta}$	firm-cl.	two-way	wild boot.	jackknife
Panel A. High-frequency monetary surprise (USMPD)							
Baseline	Market-building (log e)	Demand	1.24	0.017**	0.002***	0.021**	<.001***
		Monetary	0.57	0.474	0.392	0.373	0.411
	Productive capability (log R&D)	Demand	1.93	0.101	0.065*	0.139	0.058*
		Monetary	1.66	0.391	0.356	0.298	0.306
Gap: $\geq$ 2007	Market-building (log e)	Demand	1.24	0.016**	0.002***	0.030**	<.001***
		Monetary	0.59	0.528	0.489	0.409	0.431
	Productive capability (log R&D)	Demand	1.90	0.101	0.060*	0.144	0.061*
		Monetary	1.86	0.396	0.386	0.317	0.316
Gap: $\geq$ 2008	Market-building (log e)	Demand	1.14	0.037**	0.010***	0.092*	0.022**
		Monetary	0.39	0.671	0.638	0.614	0.674
	Productive capability (log R&D)	Demand	2.14	0.089*	0.048**	0.135	0.045**
		Monetary	2.34	0.290	0.267	0.214	0.233
Panel B. Taylor-rule residual (rule-based monetary measure)							
Baseline	Market-building (log e)	Demand	1.06	0.011**	0.003***	0.015**	0.001***
		Monetary	0.68	0.058*	0.003***	0.067*	0.121
	Productive capability (log R&D)	Demand	1.25	0.133	0.085*	0.069*	0.091*
		Monetary	0.77	0.384	0.124	0.302	0.395
Gap: $\geq$ 2007	Market-building (log e)	Demand	1.10	0.009***	0.004***	0.054*	0.022**
		Monetary	1.00	0.023**	0.036**	0.073*	0.212
	Productive capability (log R&D)	Demand	1.29	0.131	0.071*	0.134	0.141
		Monetary	1.40	0.229	0.135	0.171	0.296
Gap: $\geq$ 2008	Market-building (log e)	Demand	1.12	0.017**	0.011**	0.056*	0.031**
		Monetary	0.97	0.027**	0.031**	0.110	0.234
	Productive capability (log R&D)	Demand	1.39	0.145	0.090*	0.142	0.149
		Monetary	1.63	0.200	0.116	0.148	0.280

Notes. Each row is the coefficient on the indicated aggregate shock interacted with pre-period market-building exposure, from the joint local projection (67) at horizon $h = 2$, with firm and industry$\times$year fixed effects and shock-by-control interactions. As in Table 2, Panel A uses the high-frequency USMPD surprise and Panel B the Taylor-rule residual. Exposure is the standardized 1999–2003 firm-average market-building ratio in every row; the specifications differ only in the first shock year used in the projection (2004 in the baseline; 2007 or 2008 in the gap rows). p-values use the same four methods as Table 2: standard errors clustered by firm, two-way clustered by firm and year, the wild-cluster bootstrap on year ($B = 1999$), and the leave-one-year-out cluster jackknife. *, **, and *** denote $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

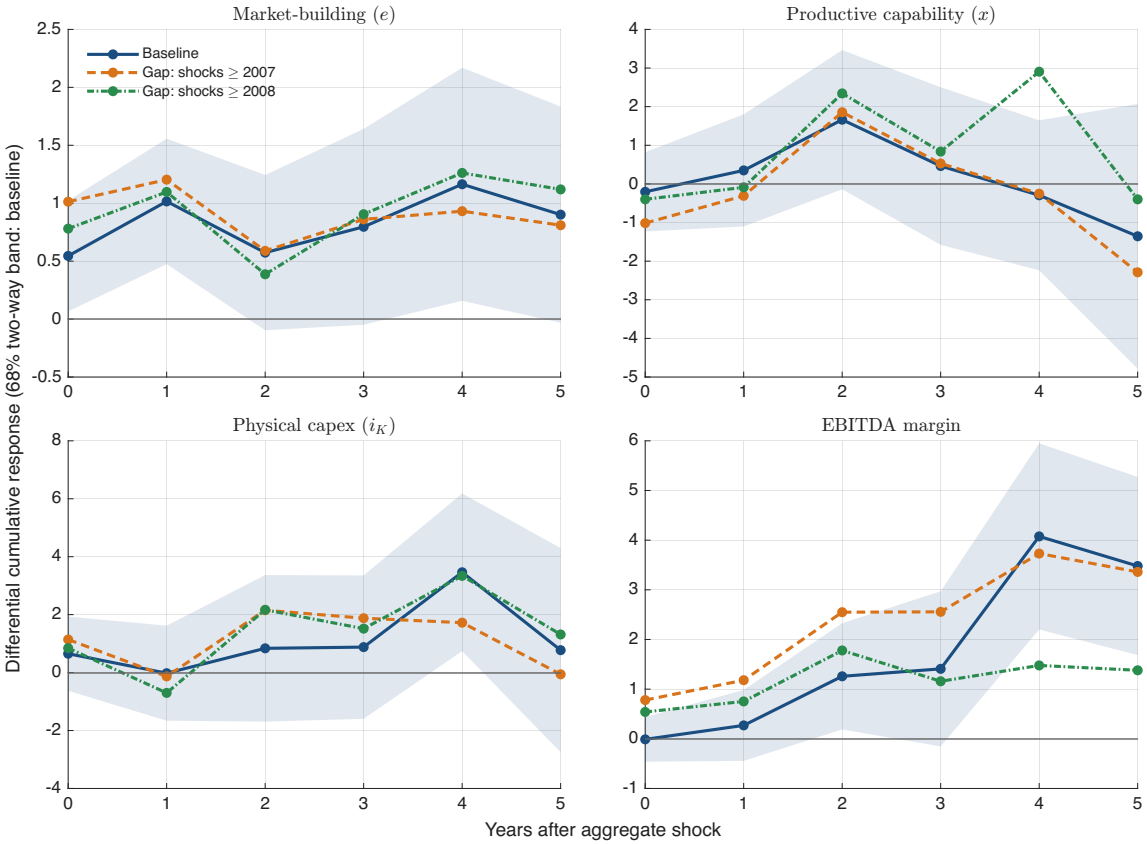
The firm-level evidence therefore supports the narrower, robust implication used in the structural model: demand expansions are the clean source of the reallocation toward customer acquisition for customer-capital-intensive firms, while the monetary transmission to the same margin is active in the model but not robustly identifiable in annual firm data. Establishing the monetary cross-sectional channel would require higher-frequency firm data or a firm-level measure of monetary exposure that varies within the year; we leave this to future work and treat the present estimates as mechanism corroboration for the demand channel. The estimates are annual cross-sectional

Figure 6: Exposure-timing robustness: responses to demand expansions



Note: The figure plots the local-projection coefficients $\beta_{D,h}$ for the four firm-level outcomes under the baseline specification and the gap-window variants that begin the shocks in 2007 and 2008. The shaded band is the baseline 68 percent two-way clustered confidence band.

Figure 7: Exposure-timing robustness: responses to monetary easing.



Note: The figure plots the local-projection coefficients $\beta_{R,h}$ for the four firm-level outcomes under the baseline specification and the gap-window variants that begin the shocks in 2007 and 2008. The shaded band is the baseline 68 percent two-way clustered confidence band.

responses and are not used to match the quarterly structural IRFs point by point.

7 Quantitative Fit

The quantitative exercise asks whether the mechanism in Proposition 3 is consistent with aggregate and firm-level moments. A positive demand disturbance raises output, inflation, the policy rate, productive-capability building, market-building, and organizational investment. The model thus delivers the central allocation margin of the paper without imposing a mechanical crowding-out restriction: expansions increase the private value of future demand power and raise both intangible margins on impact, while customer capital accumulates with a lag.

Table 4 reports impact responses for the main shocks. A demand expansion raises output by 0.894 percent, market-building by 1.227 percent, productive-capability building by 0.336 percent, and organizational investment by 14.386 percent on impact, both intangible margins therefore move in the same direction, with the demand response concentrated in market-building. Inflation and the policy rate rise, as in a standard New Keynesian expansion. The aggregate markup falls initially because real activity and labor compensation expand more than desired markups on impact, but Figure 8 shows the dynamic force emphasized by the theory: market-building rises immediately and raises the customer-capital state later in the transition.

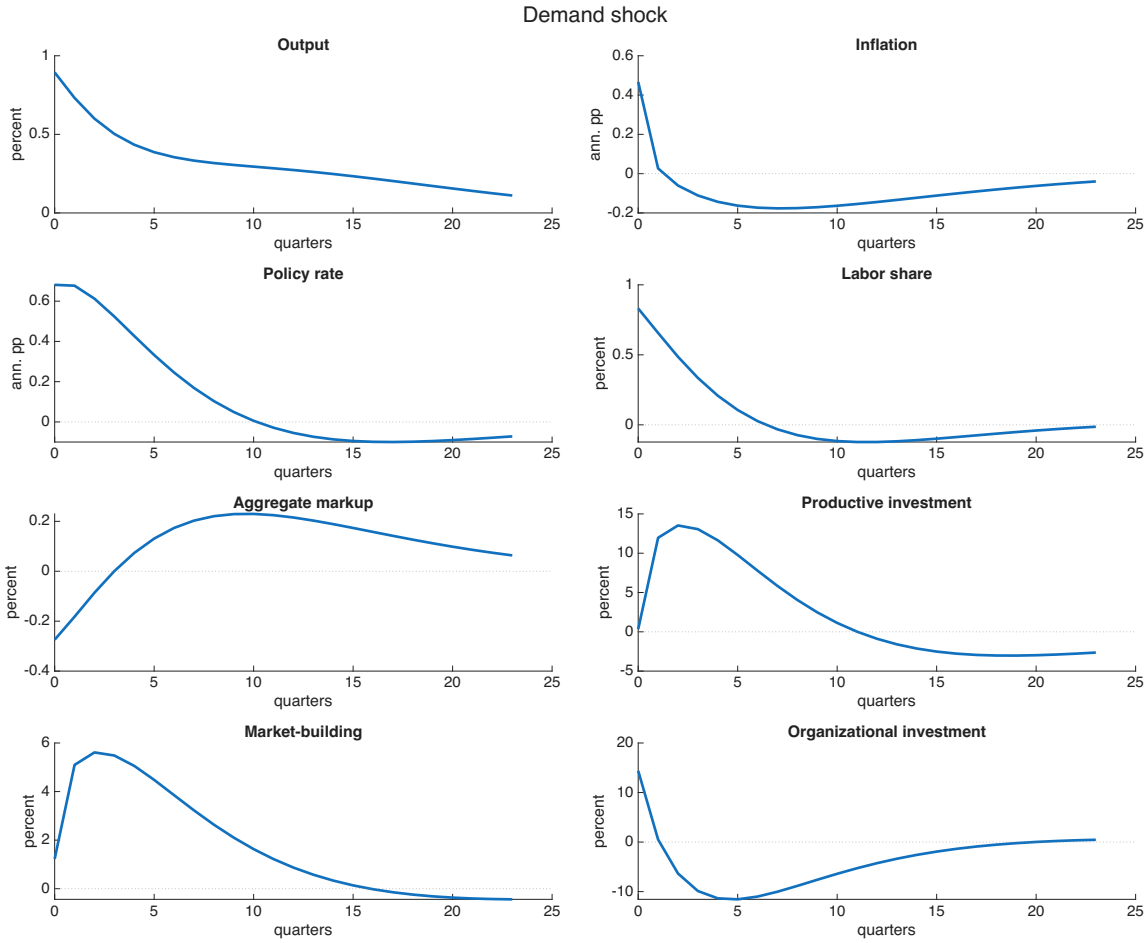
Table 4: Impact responses in the benchmark

Shock	Y	LS	μ	x	e	i^H	π	R
Demand shock	0.894	0.831	-0.274	0.336	1.227	14.386	0.467	0.681
Market-power shock	0.011	-0.003	0.002	0.221	0.146	0.624	0.035	0.024
Price-markup shock	-0.000	-0.002	0.002	-0.018	-0.016	0.089	0.008	0.004
Wage-markup shock	-0.008	-0.020	0.013	0.089	-0.015	-0.862	-0.086	-0.050
Monetary easing	0.339	0.074	-0.050	0.207	1.097	6.480	0.619	-0.114

The demand-shock impulse response has two implications. First, the customer-capital channel is expansionary on impact: higher demand raises current output and the return to acquiring customers. Second, productive implementation and customer acquisition are complementary at business-cycle frequencies in the benchmark. The firm does not merely produce more; it also invests more in the stock that makes future demand less elastic. This is the sense in which ordinary business-cycle shocks can become market-power shocks endogenously.

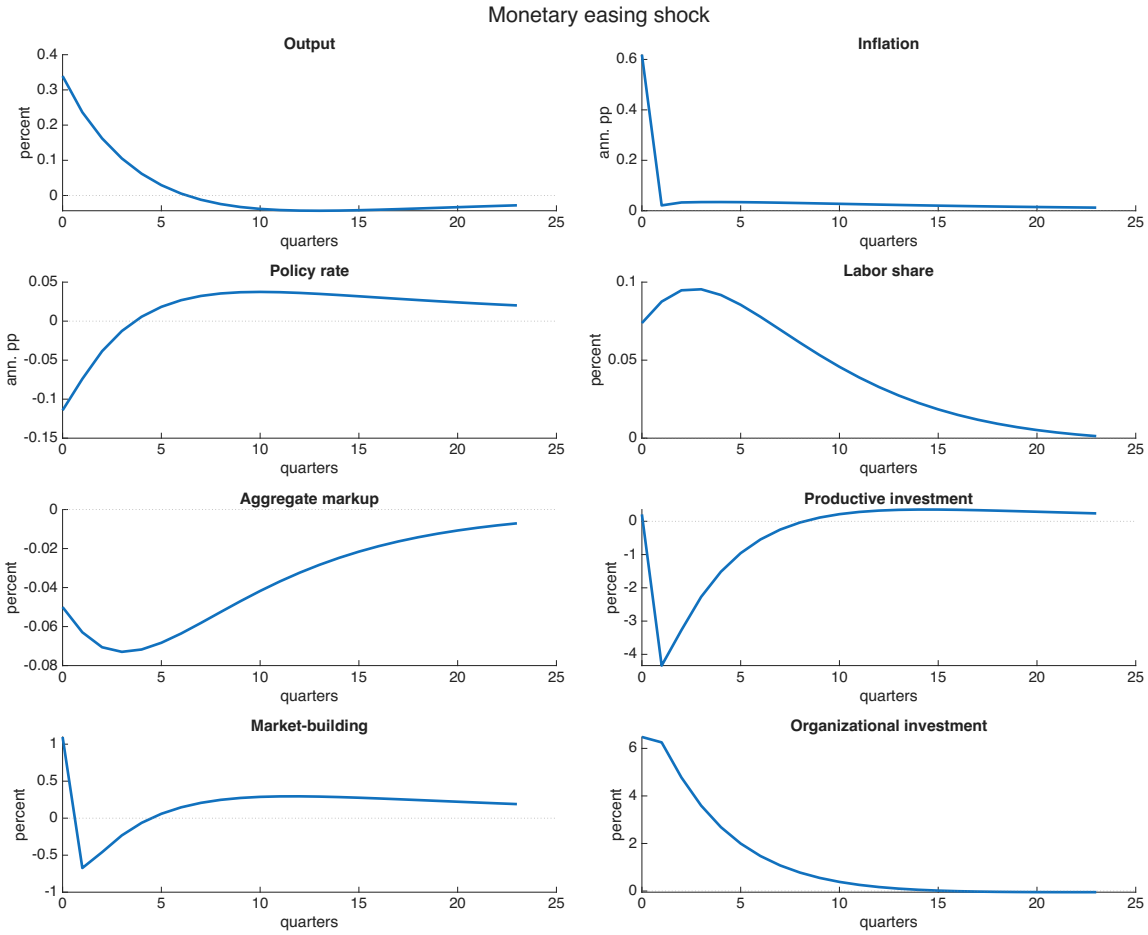
Figure 9 shows that monetary accommodation works through the same customer-acquisition margin, but indirectly. A monetary easing lowers the policy rate and raises

Figure 8: Demand-shock transmission



Note: Responses are percent deviations from steady state, except inflation and the policy rate, which are annualized percentage-point deviations.

Figure 9: Monetary-easing transmission



Note: The figure plots the response to a negative monetary-policy innovation. Responses are percent deviations from steady state, except inflation and the policy rate, which are annualized percentage-point deviations.

output, inflation, and organizational investment. On impact the easing's effect is concentrated in market-building (1.097 percent), which is nearly as large as under the demand shock and far exceeds its effect on productive-capability building (0.207 percent); productive capability then declines for several quarters before recovering. The easing therefore tilts the firm's intangible mix toward customer acquisition on impact, but demand shocks generate the larger and more persistent accumulation of customer capital. The firm-level local projections in Section 6 are cross-sectional mechanism evidence, not a point-by-point target for this aggregate quarterly IRF. Their identifying variation is the interaction of aggregate shocks with pre-period exposure; because year fixed effects absorb the aggregate shock itself, identification rests on the differential response of high-exposure firms across a limited number of aggregate years. The one differential response that is robust to year-level inference is the demand channel: customer-capital-intensive firms raise market-building after aggregate demand expansions. This corroborates the model's central mechanism in Figure 8, in which demand is the dominant and most persistent driver of market-building and customer capital. The differential response to monetary innovations carries the model-consistent positive sign under both monetary measures, but is not robustly identified under the clean high-frequency series, which is dominated by a small number of episodes, thus we report it transparently but do not lean on its magnitude. This is consistent with, rather than in tension with, the structural model: monetary accommodation moves the customer-acquisition margin only indirectly, through aggregate demand, inflation stabilization, discounting, and the financing cost of market-building, and it accounts for a secondary share of business-cycle variance. The persistent market-power accumulation channel in the calibrated model is the demand channel in Figure 8; competition policy affects the same state directly by changing the private return to customer acquisition.

Table 5 reports the implied variance decomposition. Demand shocks account for most of the variance of output, the policy rate, the labor share, and the aggregate markup. Monetary shocks account for a substantial share of inflation and output variation. By contrast, the estimated innovations to market-power, wage-markup, and price-markup wedges play little role in explaining business-cycle variation. Thus, in the benchmark, markup fluctuations are generated mainly by firms' endogenous response to demand conditions, not by large exogenous movements in desired markups. Competition-policy innovations are omitted from this decomposition because they are policy counterfactuals. Their role is evaluated in Section 8.

The selected moments in Table 6 show that the benchmark reproduces the main nominal comovements. Inflation volatility, policy-rate volatility, output–inflation comovement, and output–policy-rate comovement are close to their empirical counterparts. The largest discrepancies are real: output is somewhat too smooth and the labor share

Table 5: Variance decomposition in the benchmark

Shock	Output	Inflation	Policy rate	Labor share	Markup
Monetary	16.1	42.3	2.3	1.9	4.1
TFP	0.5	2.7	1.7	6.4	18.3
Demand	83.1	53.9	95.5	91.6	77.5
Market power	0.0	0.1	0.1	0.0	0.0
Wage markup	0.2	0.9	0.5	0.1	0.2
Price markup	0.0	0.0	0.0	0.0	0.0

Table 6: Selected moment fit

Moment	U.S. target	Benchmark	Gap
Std. inflation	0.975	1.003	+0.028
Std. policy rate	1.083	1.057	-0.026
Std. output	1.195	1.077	-0.118
Std. labor share	0.948	1.119	+0.171
Corr. y, π	0.341	0.403	+0.062
Corr. y, R	0.726	0.768	+0.042

is somewhat too volatile. Economically, this reflects the model's tight link between demand-driven sectoral reallocation and the measured aggregate labor share.

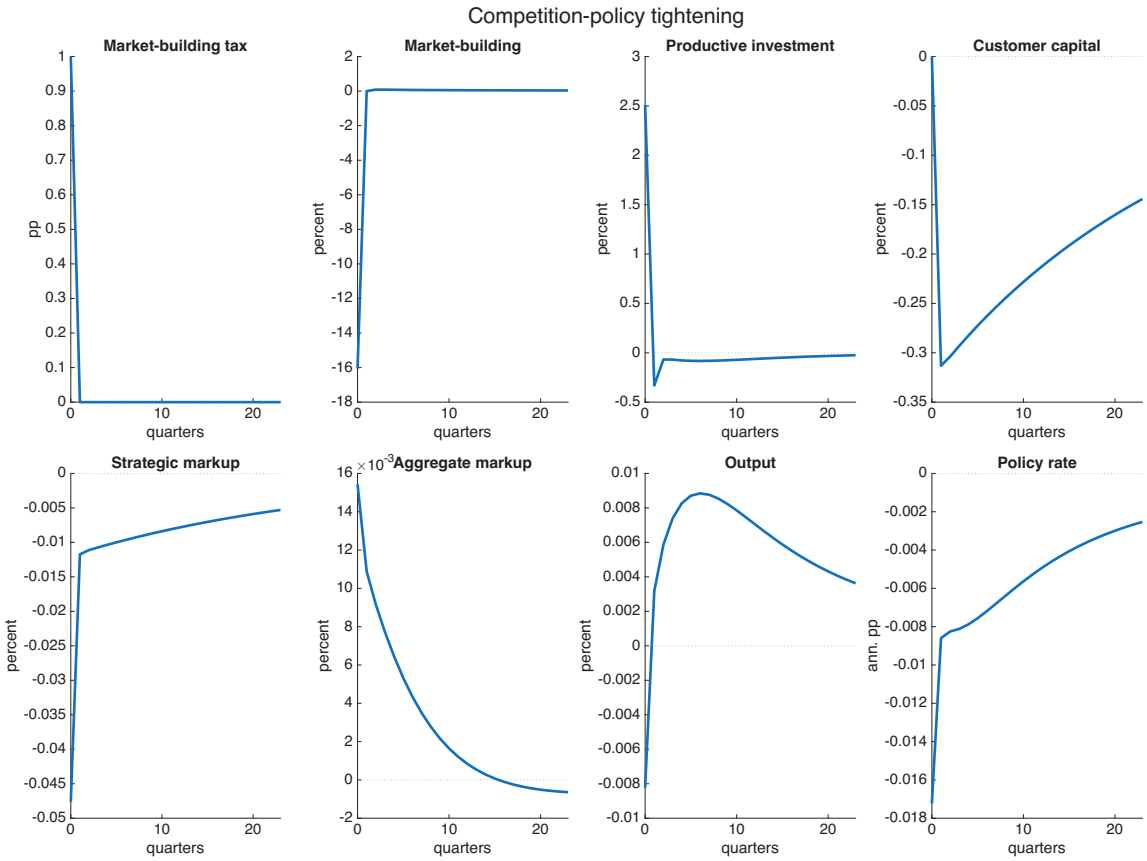
8 Policy Analysis

Proposition 4 shows the relevant wedge: firms do not internalize how customer-capital accumulation changes future markups, price-adjustment losses, and the allocation of scarce organizational resources. This creates a role for competition policy as a cyclical instrument that changes the private return to market-building. The appropriate policy experiment is therefore not to add a competition-policy shock to the variance decomposition, but to ask how the economy responds when policy raises the cost of customer acquisition.

Figure 10 reports a temporary competition-policy tightening, implemented as a one-percentage-point increase in the market-building tax $\tau_{r,t}$. The policy acts on the firm's intratemporal allocation margin. Market-building falls by 16.1 percent on impact, while productive-capability investment rises by 2.6 percent. Customer capital falls with a one-quarter lag, and the strategic-sector markup declines. The output response is small because the tax mainly reallocates intangible investment within the firm rather than operating as a broad contractionary demand shock. This is precisely the distinction between monetary and competition policy in the model: monetary policy changes

the aggregate environment in which customer acquisition is valuable, whereas competition policy changes the private price of that acquisition.

Figure 10: Competition-policy tightening in the benchmark



Note: The figure plots the response to a one-percentage-point innovation to the market-building tax $\tau_{r,t}$. Responses are percent deviations from steady state, except the tax and the policy rate, which are percentage-point and annualized percentage-point deviations, respectively.

This experiment clarifies the division of labor between the two policy instruments. Monetary policy affects market-power accumulation indirectly by moving demand, discount factors, and inflation. Competition policy affects the same dynamic margin directly by changing the private price of customer acquisition. The two policies are therefore linked but not redundant. Monetary stabilization governs the aggregate cycle; competition policy governs how much of that cycle is converted into persistent demand power.

Table 7 quantifies this division of labor using optimal simple rules. Welfare is evaluated under the calibrated six-shock stochastic process using the household utility criterion with second-order variance corrections for consumption, labor, price adjustment, wage adjustment, and regulatory resource costs. The monetary-only frontier searches over Taylor-rule coefficients with competition feedback inactive. The competition-only frontier holds the benchmark monetary rule fixed and searches over feedback

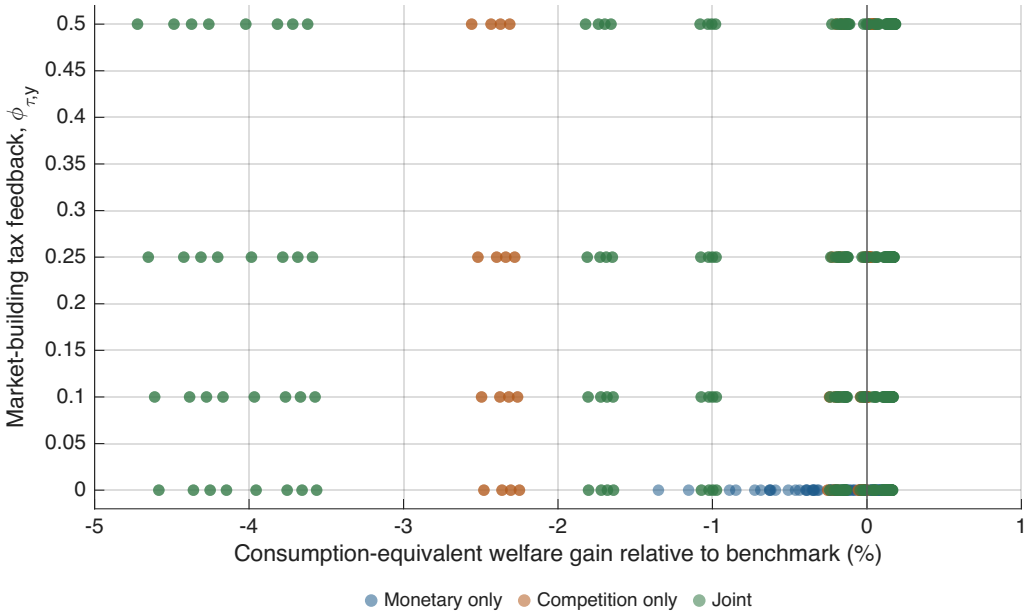
in the customer-capital depreciation and market-building tax rules. The joint frontier combines the best monetary candidates with the same competition-feedback class. The table reports consumption-equivalent welfare gains relative to the benchmark rule.

Table 7: Implementable policy counterfactuals

Regime	ϕ_π	ϕ_y	ρ_R	$\phi_{\delta,m}$	$\phi_{\tau,e}$	$\phi_{\tau,y}$	$\phi_{\tau,z}$	CE gain (%)
Benchmark rule	3.50	0.300	0.85	0.000	0.000	0.000	0.000	+0.0000
Monetary only	4.00	0.250	0.60	0.000	0.000	0.000	0.000	+0.1621
Competition only	3.50	0.300	0.85	0.000	0.000	0.500	0.050	+0.0527
Joint rule	4.00	0.250	0.60	0.020	0.000	0.500	0.050	+0.1853

The monetary-only frontier raises welfare by 0.162 percent of steady-state consumption. It does so by choosing a stronger inflation response and less interest-rate inertia than the benchmark. The competition-only frontier raises welfare by 0.053 percent, primarily by increasing the market-building tax when the natural output gap is high. The joint rule raises welfare by 0.185 percent, more than either single-instrument frontier. The gain is not a mechanical substitute for monetary stabilization: the best joint rule combines tighter nominal stabilization with a competition rule that limits the conversion of expansions into customer-capital accumulation.

Figure 11: Optimal simple-rule frontier



Note: The figure reports consumption-equivalent welfare gains relative to the benchmark across monetary-only, competition-only, and joint implementable rules.

Figure 11 and Table 8 show why competition policy is not redundant. Monetary policy can reduce welfare losses from inflation and demand fluctuations, but it cannot directly price the externality created by customer-capital accumulation. Competition

Table 8: Top-ranked rules on the implementable frontier

Frontier	Rank	ϕ_π	ϕ_y	ρ_R	$\phi_{\delta,m}$	$\phi_{\tau,e}$	$\phi_{\tau,y}$	$\phi_{\tau,z}$	CE gain (%)
Monetary frontier	1	4.00	0.250	0.60	0.000	0.000	0.000	0.000	+0.1621
Monetary frontier	2	4.00	0.375	0.60	0.000	0.000	0.000	0.000	+0.1468
Monetary frontier	3	3.50	0.250	0.60	0.000	0.000	0.000	0.000	+0.1461
Monetary frontier	4	4.00	0.125	0.60	0.000	0.000	0.000	0.000	+0.1389
Monetary frontier	5	3.00	0.250	0.60	0.000	0.000	0.000	0.000	+0.1227
Competition frontier	1	3.50	0.300	0.85	0.000	0.000	0.500	0.050	+0.0527
Competition frontier	2	3.50	0.300	0.85	0.000	0.000	0.500	0.000	+0.0526
Competition frontier	3	3.50	0.300	0.85	0.000	0.050	0.500	0.050	+0.0451
Competition frontier	4	3.50	0.300	0.85	0.000	0.050	0.500	0.000	+0.0450
Competition frontier	5	3.50	0.300	0.85	0.020	0.000	0.500	0.050	+0.0439
Joint frontier	1	4.00	0.250	0.60	0.020	0.000	0.500	0.050	+0.1853
Joint frontier	2	4.00	0.250	0.60	0.020	0.000	0.500	0.000	+0.1852
Joint frontier	3	4.00	0.250	0.60	0.020	0.050	0.500	0.050	+0.1823
Joint frontier	4	4.00	0.250	0.60	0.020	0.050	0.500	0.000	+0.1822
Joint frontier	5	4.00	0.250	0.60	0.000	0.000	0.500	0.050	+0.1810

policy is most valuable in states where the cyclical value of customer acquisition is high. The joint rule exploits both margins: monetary policy stabilizes the aggregate environment, while competition policy changes the private return to building persistent demand power. Within the implementable-rule class, the two instruments are therefore quantitative complements.

9 Conclusion

This paper develops a tractable mechanism linking monetary policy, intangible investment, and future market power. When firms can build customer capital, monetary conditions affect more than current demand. They also affect the relative payoff to market-building versus productive capability building. This creates a channel through which stabilization policy can shape the future markup distribution.

The structural calibration shows that this mechanism can be disciplined by firm-level data and U.S. business-cycle moments. In the model, demand expansions raise output, organizational investment, and customer capital more persistently than monetary innovations. Firm-level evidence supports the underlying demand margin: customer-capital-intensive firms robustly raise market-building after aggregate demand expansions, while the cross-sectional monetary response is positive under both monetary measures but less robustly identified under the clean high-frequency measure. Competition policy directly restrains customer acquisition and the strategic markup. The

variance decomposition reinforces the interpretation: business-cycle markup movements arise primarily from endogenous demand-to-market-power transmission, not from large exogenous markup shocks. Monetary policy stabilizes aggregate demand and inflation, while competition policy limits the conversion of expansions into durable demand power. The implementable-rule frontiers show that the joint use of the two instruments delivers larger welfare gains than either instrument alone.

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Appendix

This appendix provides supporting material for the model, calibration, empirical implementation, and analytical results. Appendix A defines the firm-level objects used to discipline the customer-capital and organizational-throughput margins and reports the full calibration. Appendix B records the full equilibrium system. Appendix C contains the proofs of the propositions and corollaries in the main text.

A Measurement and Calibration Details

This appendix reports the empirical objects used to discipline the customer-capital and organizational-throughput margins, together with the full benchmark calibration.

Firm-level variables

Table 9 maps the model's new state variables into firm-level accounting objects. The local-projection counterpart of productive capability building is R&D; broader productive investment, including capital expenditure, is reported separately as a descriptive accounting object because physical capital has its own accumulation block in the model. In the baseline firm-year panel, market-building is measured as SG&A net of R&D, scaled by sales, with gross SG&A and advertising used as alternative measures of the same customer-acquisition margin. Customer capital is constructed as a persistent stock accumulated from market-building expenditures. Markups and implied elasticities discipline the mapping from customer capital into demand power.

Table 9: Firm-level variables and construction of model objects

Model object	Accounting fields	Construction	Economic interpretation	Disciplined parameters
Strategic-sector revenue share	Sales by firm and industry	Sum sales of customer-capital-intensive firms and divide by aggregate sample sales after standard public-firm filters.	Pins down the steady-state scale of the strategic sector in final demand.	s_S^{ss}, Ω_S
Productive capability building x_{it}	R&D, sales, assets; capital expenditures reported separately	Use R&D as the clean empirical counterpart of productive capability in the local projections; report R&D plus capital expenditures as a broader descriptive investment measure.	Measures resources devoted to future efficiency and productive capability, while keeping physical capital distinct from the model's intangible capability stock.	η_A, ξ_x, δ_A
Market-building effort e_{it}	Sales, COGS, EBITDA, R&D; SG&A and advertising where available	The baseline measure is SG&A net of R&D, scaled by sales, with gross SG&A and advertising as narrower alternatives.	Measures resources devoted to selling, marketing, distribution, and customer acquisition; broader than advertising alone, it is interpreted as a proxy for market-building effort.	$\eta_M, \xi_e, \kappa_e, \omega_e$
Customer capital $\bar{m}_{it}$	Market-building effort, sales growth, markup proxies	Build a perpetual-inventory stock from market-building effort and estimate persistence from firm-level transitions.	Persistent demand power inherited from past market-building.	$\delta_m^{ss}, \eta_M, \xi_e$

Model object	Accounting fields	Construction	Economic interpretation	Disciplined parameters
Demand elasticity and markups	Sales, cost of goods sold, operating expenses, capital, employment	Estimate firm markups from production-cost data; convert markups into implied elasticities using $\varepsilon_{it} = \mu_{it}/(\mu_{it} - 1)$.	Disciplines how customer capital maps into desired markups.	$\bar{\varepsilon}, \phi_m, \phi_{zM}$
Organizational throughput	Intangible activity, operating-cost proxies, employment, managerial compensation proxies	Relate total intangible activity $x_{it} + \omega_e e_{it}$ to organizational assets and overhead intensity.	Measures the scarcity of internal resources that jointly support productive and market-building investment.	ϕ_O, ξ_O, ψ_O, Ψ_H
Labor share and operating scale	Payroll where available, employment, operating expenses, sales	Combine aggregate labor-share targets with firm-level labor and operating-cost measures to discipline steady-state labor compensation.	Anchors the production side and prevents market-power parameters from absorbing labor-share variation.	χ, LS^{ss}, LS^{alt}

Table 10: Firm-level measurement summary

Variable	Unit	Mean	Std. dev.	P25	P50	P75	Obs.
Sales	millions of dollars	3431.86	17484.40	49.15	300.60	1482.52	73008
Broad productive investment (R&D+capex)	percent of sales	13.90	20.71	2.41	6.01	15.97	72814
Market-building proxy	percent of sales	30.02	28.12	12.52	22.55	36.38	71352
Gross margin	percent of sales	38.39	23.58	23.11	36.30	53.68	73008
EBITDA margin	percent of sales	4.29	29.62	1.89	9.33	17.37	72859
Panel coverage	1999–2024	7134 firms	26 years				73008

Calibration

Table 11 summarizes the source of identification by block. Table 12 reports the full benchmark calibration. Parameters fixed at conventional macro values are separated from steady-state accounting targets, internally derived steady-state objects, normalizations, restrictions, and parameters fitted to aggregate business-cycle moments.

Table 12: Benchmark calibration

Parameter	Interpretation	Value	Discipline	Target or rationale
β	Discount factor	0.99	Standard	Quarterly real rate.
σ	Risk aversion	1.50	Standard	Medium-scale New Keynesian benchmark.
h	External habit	0.65	Standard	Consumption persistence.
φ	Inverse Frisch elasticity	5.00	Targeted standard	Labor supply curvature and labor-share target.
χ	Labor-disutility weight	1.00	Normalization	Normalizes steady-state labor units.
$\bar{N}$	Strategic-sector overhead labor	0.00	Restriction	No fixed overhead in benchmark.
α	Strategic-sector capital weight	0.25	Sectoral target	Allows separate intangible accumulation outside physical capital.

Parameter	Interpretation	Value	Discipline	Target or rationale
ρ_{KN}	Strategic-sector capital-labor substitution parameter	-0.40	Shape restriction	Smooth substitution in strategic production.
α_O	Outside-sector capital weight	0.4375	Derived SS	Chosen jointly with outside productivity to match sectoral allocation.
A_O	Outside-sector productivity	0.3082	Derived SS	Backs out the steady-state outside-sector scale.
δ_K	Physical-capital depreciation	0.025	Standard	Quarterly depreciation.
$\phi_{K,S}$	Strategic-sector investment adjustment cost	5.0	Standard	Smooths physical investment.
$\phi_{K,O}$	Outside-sector investment adjustment cost	40.0	Sectoral target	Stabilizes sectoral capital allocation.
η_Y	Final-good substitution elasticity	3.0	Sectoral target	Sectoral substitution and revenue-share fit.
s_S^{ss}	Strategic-sector revenue share	0.302	Firm-level target	Aggregate sales share of strategic sector.
Ω_S	Strategic-sector demand weight	0.4079	Derived SS	Implied by the strategic-sector revenue-share target.
$\bar{\varepsilon}$	Baseline demand elasticity	8.0	Markup target	Strategic-sector markup near 1.19 in steady state.
ϕ_m	Elasticity sensitivity to customer capital	2.20	Markup target	Markup level and cyclicality.
ϕ_{zM}	Market-power efficiency loading	2.00	Propagation	Identifies autonomous market-power disturbances.
$\phi_{\Omega B}$	Demand composition loading	2.00	Moment-fitted	Cyclical movement in strategic-sector demand and labor-share volatility.
$\phi_{\Omega M}$	Market-power composition loading	0.00	Restriction	Benchmark excludes direct composition effect of z_t^M.
Γ_M	Baseline customer-capital dividend	0.10	SS target	Keeps customer capital valuable in steady state.
Λ_M	Demand loading in customer value	0.00	Restriction	Benchmark disciplines demand through composition and current demand.
Λ_{ZM}	Private rent loading of z_t^M	0.00	Restriction	Excludes direct rent transfer from market-power shock.
υ_M	Output-gap loading in customer value	0.20	Propagation	Makes customers more valuable in booms.
ζ_e	Current demand effect of market-building	0.35	Propagation	Impact relevance of market-building around steady state.
E^{ss}	Steady-state market-building effort	0.2807	Derived SS	Implied by the steady-state customer-capital law.
δ_A	Productive-capability depreciation	0.025	Standard	Quarterly persistence.
δ_m^{ss}	Customer-capital depreciation	0.030	Persistence target	Persistence of demand power.
δ_H	Organizational-capital depreciation	0.040	Persistence target	Persistence of internal organizational resources.
η_A	Productive-capability efficiency	0.0108	SS FOC	Steady-state resource allocation.
ϕ_{AX}	Market-size elasticity of productive implementation	0.05	Propagation	Modest implementation accelerator for productive capability.
η_M	Customer-capital efficiency	0.0850	SS FOC	Steady-state market-building intensity.
κ_e	Private operating cost of market-building	0.105	SS FOC	Market-building first-order condition.
ω_e	Organizational intensity of market-building	1.35	Firm-level target	Relative internal-resource use of e_t.
ξ_x	Curvature of productive-capability building	0.70	Shape restriction	Diminishing returns in x_t.
ξ_e	Curvature of market-building	0.65	Shape restriction	Diminishing returns in e_t.
ξ_O	Throughput curvature	0.30	Propagation	Convex organizational throughput cost.
ψ_O	Organizational-stock elasticity	2.00	Propagation	Strength of inherited organizational capacity.
ϕ_O	Throughput cost scale	0.2063	Derived SS	Implied by steady-state marginal throughput cost.

Parameter	Interpretation	Value	Discipline	Target or rationale
Ψ_H	Organizational-investment convex cost	0.0075	Propagation	Smooths organizational investment.
ξ_H	Organizational-investment curvature	1.00	Shape restriction	Quadratic marginal cost in organizational investment.
Ψ_X	Productive-building flow cost	0.00	Restriction	Benchmark uses shared throughput cost.
Ψ_E	Market-building flow cost	0.00	Restriction	Benchmark uses shared throughput cost.
θ	Price adjustment cost	450	Nominal propagation	Inflation volatility and persistence.
γ_π	Price-indexation coefficient	0.00	Restriction	Benchmark uses current inflation in pricing and policy.
θ_w	Wage adjustment cost	800	Nominal propagation	Wage rigidity.
ε_w	Wage elasticity	6	Standard	Wage markup around 20 percent.
ϕ_P	Price-markup shock loading	1.00	Normalization	Shock scale absorbed by σ_P.
D_L	Market-building finance duration	20 quarters	Standard	Five-year effective duration for financed customer acquisition.
κ_L	Long-claim coupon parameter	0.9596	Derived	Implied by D_L and β.
Q_L^{ss}	Long-claim steady-state price	19.800	Derived	Steady-state price of the financing claim.
$\phi_{fin,e}$	Finance exposure of market-building	0.15	Moment-fitted	Monetary transmission into customer-acquisition costs.
$\phi_{fin,x}$	Finance exposure of productive capability	0.00	Restriction	Productive-capability financing channel inactive in benchmark.
ρ_e^s	Persistence of market-building spread	0.95	Standard	Persistent financing conditions.
χ_R^s	Monetary-shock loading of spread	500	Propagation	Maps policy innovations into the effective customer-acquisition cost.
ρ_R	Interest-rate smoothing	0.85	Standard	Taylor-rule inertia.
ϕ_π	Inflation response	3.50	Moment-fitted	Nominal comovement and Taylor principle.
ϕ_y	Output-cycle response	0.30	Moment-fitted	Response to observed output cycle.
$\phi_{\Delta y}$	Output-growth response	0.50	Moment-fitted	Speed-limit component of the policy rule.
π^{ss}	Gross steady-state inflation	1.00	Normalization	Zero steady-state inflation.
ρ_δ	Competition-rule persistence	0.90	Policy benchmark	Institutional persistence in customer-capital depreciation policy.
$\phi_{\delta,m}$	Depreciation response to customer-capital gap	0.00	Restriction	Cyclical depreciation rule inactive in the benchmark.
$\phi_{\tau,e}$	Tax response to market-building gap	0.00	Restriction	Cyclical tax rule inactive in the benchmark.
$\phi_{\tau,y}$	Tax response to output gap	0.00	Restriction	Policy analysis varies this coefficient away from benchmark.
$\phi_{\tau,z}$	Tax response to market-power shock	0.00	Restriction	Policy analysis varies this coefficient away from benchmark.
τ_r^{ss}	Steady-state market-building tax	0.00	Restriction	No long-run tax in the benchmark.
δ_m^{ref}	Reference customer-capital depreciation	0.030	Derived	Set equal to benchmark customer-capital depreciation.
τ_r^{ref}	Reference long-run tax	0.00	Restriction	No long-run tax benchmark.
ω_δ	Cyclical depreciation-rule cost	0.40	Normative	Resource cost of using depreciation policy.
ω_τ	Cyclical tax-rule cost	0.40	Normative	Resource cost of using tax policy.
$\bar{\omega}_\delta$	Long-run depreciation-stance cost	600	Normative	Penalizes permanent stance changes.
$\bar{\omega}_\tau$	Long-run tax-stance cost	120	Normative	Penalizes permanent stance changes.
ρ_A	TFP persistence	0.95	Standard	Persistent technology.
ρ_B	Demand-shock persistence	0.85	Shock persistence	Business-cycle persistence.

Parameter	Interpretation	Value	Discipline	Target or rationale
ρ_M	Market-power-shock persistence	0.85	Shock persistence	Persistent private market-power disturbances.
ρ_W	Wage-markup-shock persistence	0.85	Shock persistence	Persistent wage markup wedge.
ρ_P	Price-markup-shock persistence	0.85	Shock persistence	Persistent cost-push wedge.
σ_R	Monetary shock s.d.	0.0015	Moment-fitted	Policy-rate innovations.
σ_A	TFP shock s.d.	0.0050	Moment-fitted	Annual TFP-growth volatility.
σ_B	Demand shock s.d.	0.0421	Moment-fitted	GDP cycle and macro comovement.
σ_M	Market-power shock s.d.	0.0150	Moment-fitted	Markup volatility orthogonal to output and inflation.
σ_W	Wage-markup shock s.d.	0.0200	Moment-fitted	Wage markup wedge.
σ_P	Price-markup shock s.d.	0.0062	Moment-fitted	Inflation volatility and output–inflation comovement.

B Equilibrium System

This appendix records the equilibrium system used in the quantitative analysis. The notation matches the model section.

Actual allocation

Household block.

$$\lambda_t = \exp(z_t^B)(C_t - hC_{t-1})^{-\sigma} - \beta h\, \mathbb{E}_t[\exp(z_{t+1}^B)(C_{t+1} - hC_t)^{-\sigma}], \quad (70)$$

$$1 = \beta \mathbb{E}_t[\lambda_{t+1} R_t / (\lambda_t \pi_{t+1})], \quad (71)$$

$$\pi_{w,t} = \pi_t w_t / w_{t-1}, \quad (72)$$

$$\theta_w \pi_{w,t}(\pi_{w,t} - 1) = \varepsilon_w \chi N_t^{\varphi} / (\lambda_t w_t) - (\varepsilon_w - 1)\exp(z_t^W) + \theta_w \beta \mathbb{E}_t[\lambda_{t+1} \pi_{w,t+1}(\pi_{w,t+1} - 1) w_{t+1} N_{t+1} / (\lambda_t w_t N_t)]\,. \quad (73)$$

Table 11: Calibration discipline by block

Block	Parameters	Discipline
External macro block	$\beta, \sigma, h, \delta_K, \phi_{K,S}$ $\varepsilon_w, \rho_R, \phi_\pi, \phi_y, \phi_{\Delta y}, \rho_A$	Fixed at standard quarterly values or conventional policy-rule values. These parameters are not used to fit the customer-capital mechanism.
Accounting and steady-state block	$s_S^{ss}, \Omega_S, A_O, \alpha_O$ $\bar{\varepsilon}, \phi_m, \eta_A, \eta_M$ $\kappa_e, \omega_e, \chi, \phi_O$	Pinned by sectoral revenue share, aggregate labor share, the strategic-sector markup, intangible intensity, and steady-state first-order conditions.
Exclusion restrictions and normalizations	$\phi_{\Omega M}, \Lambda_M, \Lambda_{ZM}$ $\Psi_X, \Psi_E, \phi_{\delta,m}$ $\phi_{\tau,e}, \phi_{\tau,y}, \phi_{\tau,z}, \tau_r^{ss}$	Set to zero in the benchmark to avoid fitting the core result with direct rent transfers or active competition-policy feedback.
Moment-fitted shock scales	$\sigma_R, \sigma_A, \sigma_B, \sigma_M, \sigma_W, \sigma_P$	Chosen to match business-cycle volatilities and co-movements; these are shock scales rather than preference or technology primitives.
Propagation parameters	$\theta, \theta_w, \gamma_\pi, \phi_{\Omega B}, \phi_{zM}$ $\zeta_e, \upsilon_M, \phi_{AX}, \phi_{fin,e}$ $D_L, \rho_e^s, \chi_R^s, \xi_O, \psi_O, \Psi_H$	Discipline nominal propagation, demand-to-customer-capital transmission, productive implementation, and the financing exposure of market-building.

Final demand and sectors.

$$\Omega_{S,t} = \left[1 + \exp\left\{-\left(\log\frac{\Omega_S}{1-\Omega_S} + \phi_{\Omega B} z_t^B + \phi_{\Omega M} z_t^M\right)\right\}\right]^{-1}, \tag{74}$$

$$Y_{S,t} = \Omega_{S,t} p_{S,t}^{-\eta_Y} Y_t, \tag{75}$$

$$Y_{O,t} = (1-\Omega_{S,t}) p_{O,t}^{-\eta_Y} Y_t, \tag{76}$$

$$1 = \Omega_{S,t} p_{S,t}^{1-\eta_Y} + (1-\Omega_{S,t}) p_{O,t}^{1-\eta_Y}, \tag{77}$$

$$Y_{S,t} = \exp(z_t^A)\overline{A}_t \left[\alpha K_{S,t-1}^{\rho_{KN}} + (1-\alpha)(N_{S,t} - \bar{N})^{\rho_{KN}}\right]^{1/\rho_{KN}}, \tag{78}$$

$$Y_{O,t} = A_O K_{O,t-1}^{\alpha_O} N_{O,t}^{1-\alpha_O}, \tag{79}$$

$$N_t = N_{S,t} + N_{O,t}. \tag{80}$$

Factor-price conditions are those in Equations (13)–(17).

Elasticity, pricing, and markups.

$$d_t = 1 + \bar{m}_t + \zeta_e(e_t - E^{ss}), \tag{81}$$

$$\varepsilon_t = \bar{\varepsilon} - \phi_m \exp(\phi_{zM} z_t^M) \log d_t, \tag{82}$$

$$\mu_t^* = \varepsilon_t/(\varepsilon_t - 1), \tag{83}$$

$$\theta\pi_{S,t}(\pi_{S,t}-1) = 1 - \varepsilon_t + \varepsilon_t mc_t + \phi_P z_t^P + \theta\mathbb{E}_t[\mathrm{disc}_{t,t+1}(Y_{S,t+1}/Y_{S,t})\pi_{S,t+1}(\pi_{S,t+1}-1)], \tag{84}$$

$$\pi_{S,t} = \pi_t p_{S,t}/p_{S,t-1}, \tag{85}$$

$$\mu_t^{agg} = Y_t/(mc_t p_{S,t} Y_{S,t} + p_{O,t} Y_{O,t}), \tag{86}$$

$$LS_t = w_t N_t / Y_t. \tag{87}$$

Shadow values and intangible choices.

$$q_t^L = \mathbb{E}_t[\text{disc}_{t,t+1}(1+\kappa_L q_{t+1}^L)], \tag{88}$$

$$s_t^e = \rho_e^s s_{t-1}^e + \chi_R^s \varepsilon_t^R, \tag{89}$$

$$\mathcal{F}_t^e = \left(\frac{Q_L^{ss}}{q_t^L}\right)^{\phi_{fin,e}} \exp(\phi_{fin,e} D_L s_t^e/400), \tag{90}$$

$$\mathcal{F}_t^x = \left(\frac{Q_L^{ss}}{q_t^L}\right)^{\phi_{fin,x}} \exp(\phi_{fin,x} D_L s_t^e/400), \tag{91}$$

$$q_t^A = p_{S,t} Y_{S,t} \frac{\mu_t^* - 1}{\mu_t^*} \frac{1}{\overline{A}_t} + \mathbb{E}_t[\text{disc}_{t,t+1}(1-\delta_A) q_{t+1}^A], \tag{92}$$

$$q_t^B = \mathbb{E}_t[\text{disc}_{t,t+1}\, q_{t+1}^A], \tag{93}$$

$$q_t^P = q_t^A, \tag{94}$$

$$q_t^m = p_{S,t} Y_{S,t} \left(\Gamma_M + \Lambda_M z_t^B + \Lambda_{ZM} z_t^M + \frac{\phi_m}{d_t \varepsilon_t^2} + v_M gap_t\right) + \mathbb{E}_t[\text{disc}_{t,t+1}(1-\delta_{m,t}) q_{t+1}^m], \tag{95}$$

$$T_t = x_t + \omega_e e_t, \tag{96}$$

$$\Gamma_t = \phi_O T_t^{1+\xi_O} H_{t-1}^{-\psi_O}/(1+\xi_O), \tag{97}$$

$$\nu_t^H = \phi_O T_t^{\xi_O} H_{t-1}^{-\psi_O}, \tag{98}$$

$$C_t^E = \Psi_E e_t^{1+\xi_e}/(1+\xi_e), \tag{99}$$

$$C_t^H = \Psi_H (i_t^H)^{1+\xi_H}/(1+\xi_H), \tag{100}$$

$$C_t^X = \Psi_X x_t^{1+\xi_x}/(1+\xi_x), \tag{101}$$

$$\mathcal{S}_t^A = (Y_{S,t}/Y_S^{ss})^{\phi_{AX}}, \tag{102}$$

$$\eta_A \xi_x x_t^{\xi_x - 1} \mathcal{S}_t^A q_t^P = \mathcal{F}_t^x \nu_t^H + \Psi_X x_t^{\xi_x}, \tag{103}$$

$$\eta_M \xi_e e_t^{\xi_e - 1} \mathbb{E}_t[\text{disc}_{t,t+1}\, q_{t+1}^m] + \zeta_e p_{S,t} Y_{S,t} \phi_m/(d_t \varepsilon_t^2) = \omega_e \nu_t^H + \mathcal{F}_t^e \kappa_e + \tau_{r,t} + \Psi_E e_t^{\xi_e}, \tag{104}$$

$$1 + \Psi_H (i_t^H)^{\xi_H} = \mathbb{E}_t\Big[\text{disc}_{t,t+1}\Big(\psi_O \phi_O T_{t+1}^{1+\xi_O} H_t^{-\psi_O - 1}/(1+\xi_O) + (1-\delta_H)\{1 + \Psi_H (i_{t+1}^H)^{\xi_H}\}\Big)\Big], \tag{105}$$

$$H_{t+1} = (1-\delta_H) H_t + i_t^H, \tag{106}$$

$$A_t^{build} = \eta_A x_t^{\xi_x} \mathcal{S}_t^A, \tag{107}$$

$$\overline{A}_t = (1-\delta_A)\overline{A}_{t-1} + A_t^{build}, \tag{108}$$

$$\bar{m}_t = (1-\delta_{m,t-1})\bar{m}_{t-1} + \eta_M e_{t-1}^{\xi_e}. \tag{109}$$

Sectoral physical capital.

$$K_{j,t} = (1-\delta_K)K_{j,t-1} + i^K_{j,t}, \qquad j \in \{S,O\}, \tag{110}$$

$$Q^K_{j,t} = 1 + \phi_{K,j}(i^K_{j,t}/K_{j,t-1} - \delta_K), \tag{111}$$

$$\begin{aligned} Q^K_{j,t} = \mathbb{E}_t\Big[\mathrm{disc}_{t,t+1}\Big(r^k_{j,t+1} + (1-\delta_K)Q^K_{j,t+1} \\ +\phi_{K,j}\left(\frac{i^K_{j,t+1}}{K_{j,t}} - \delta_K\right)\frac{i^K_{j,t+1}}{K_{j,t}} - \frac{\phi_{K,j}}{2}\left(\frac{i^K_{j,t+1}}{K_{j,t}} - \delta_K\right)^2\Big)\Big]. \end{aligned} \tag{112}$$

Policy and resource feasibility.

$$z^A_t = \rho_A z^A_{t-1} + \varepsilon^A_t, \tag{113}$$

$$z^B_t = \rho_B z^B_{t-1} + \varepsilon^B_t, \tag{114}$$

$$z^M_t = \rho_M z^M_{t-1} + \varepsilon^M_t, \tag{115}$$

$$z^W_t = \rho_W z^W_{t-1} + \varepsilon^W_t, \tag{116}$$

$$z^P_t = \rho_P z^P_{t-1} + \varepsilon^P_t, \tag{117}$$

$$\frac{R_t}{R^{ss}} = \left(\frac{R_{t-1}}{R^{ss}}\right)^{\rho_R}\left[\left(\frac{\pi_t}{\pi^{ss}}\right)^{\phi_\pi}\exp\{\Xi^R_t\}\right]^{1-\rho_R}\exp(\varepsilon^R_t), \tag{118}$$

$$\Xi^R_t = \phi_y(\log Y_t - \log Y^{ss}) + \phi_{\Delta y}(\log Y_t - \log Y_{t-1}), \tag{119}$$

$$\delta_{m,t} = (1-\rho_\delta)\delta^{ss}_m + \rho_\delta\delta_{m,t-1} + \phi_{\delta,m}(\bar{m}_t - \bar{m}^{nat}_t) + \varepsilon^\delta_t, \tag{120}$$

$$\tau_{r,t} = \tau^{ss}_r + \phi_{\tau,e}(e_t - e^{nat}_t) + \phi_{\tau,y}gap_t + \phi_{\tau,z}z^M_t + \varepsilon^\tau_t, \tag{121}$$

$$gap_t = \log Y_t - \log Y^{nat}_t, \tag{122}$$

$$Y_t = C_t + i^K_{S,t} + i^K_{O,t} + i^H_t + \Gamma_t \tag{123}$$

$$+ C^H_t + C^X_t + C^E_t \tag{124}$$

$$+ K_{S,t-1}\frac{\phi_{K,S}}{2}\left(\frac{i^K_{S,t}}{K_{S,t-1}} - \delta_K\right)^2 \tag{125}$$

$$+ K_{O,t-1}\frac{\phi_{K,O}}{2}\left(\frac{i^K_{O,t}}{K_{O,t-1}} - \delta_K\right)^2 \tag{126}$$

$$+ \frac{\theta}{2}(\pi_{S,t} - 1)^2 p_{S,t}Y_{S,t} \tag{127}$$

$$+ \frac{\theta_w}{2}(\pi_{w,t} - 1)^2 w_t N_t \tag{128}$$

$$+ G^{reg}_t. \tag{129}$$

Natural allocation

The natural block duplicates the real side under flexible prices and has its own variables

$$\{Y_t^{nat}, Y_{S,t}^{nat}, Y_{O,t}^{nat}, C_t^{nat}, N_t^{nat}, K_{S,t}^{nat}, K_{O,t}^{nat}, \overline{A}_t^{nat}, \bar{m}_t^{nat},$$
$$x_t^{nat}, e_t^{nat}, H_t^{nat}, q_t^{A,nat}, q_t^{B,nat}, q_t^{P,nat}, q_t^{m,nat}, q_t^{L,nat}\}.$$

It uses the same smooth-throughput and accumulation equations, omits the Rotemberg price-adjustment terms, keeps the multiplicative z_t^M elasticity schedule, omits the $\Lambda_{ZM} z_t^M$ rent loading, and sets the natural gap used in the firm-value and competition-policy blocks. The natural block also uses $q_t^{P,nat} = q_t^{A,nat}$, $\mathcal{S}_t^{A,nat} = (Y_{S,t}^{nat}/Y_S^{ss})^{\phi_{AX}}$, and the same long-duration financing wedge with no monetary-policy innovation in the spread process.

C Proofs

Proof of Proposition 1

Let

$$F_x(x; q, \nu^x, \Psi_X) = \eta_A \xi_x x^{\xi_x - 1} q - \nu^x - \Psi_X x^{\xi_x}.$$

At an interior solution,

$$F_{x,x} = \eta_A \xi_x (\xi_x - 1) x^{\xi_x - 2} q - \Psi_X \xi_x x^{\xi_x - 1} < 0$$

because $0 < \xi_x < 1$ and $\Psi_X \geq 0$. The implicit-function theorem gives a unique local mapping. Since $F_{x,q} > 0$, $F_{x,\nu^x} < 0$, and $F_{x,\Psi_X} < 0$, the signs are $X_q = -F_{x,q}/F_{x,x} > 0$, $X_{\nu^x} = -F_{x,\nu^x}/F_{x,x} < 0$, and $X_{\Psi_X} = -F_{x,\Psi_X}/F_{x,x} < 0$.

For market-building, define

$$F_e(e; \widetilde{q}, \mathcal{D}, \nu, \tau, \omega_e, \mathcal{F}^e) = \eta_M \xi_e e^{\xi_e - 1} \widetilde{q} + \mathcal{D} - \omega_e \nu - \mathcal{F}^e \kappa_e - \tau - \Psi_E e^{\xi_e}.$$

Condition (65) is $F_{e,e} < 0$. The implicit-function theorem then delivers the local mapping. Because $F_{\widetilde{q}} > 0$, $F_{\mathcal{D}} > 0$, $F_\nu < 0$, $F_\tau < 0$, and $F_{\omega_e} < 0$, the stated signs follow. If $\kappa_e > 0$, $F_{\mathcal{F}^e} = -\kappa_e < 0$, so a higher financing wedge lowers market-building. □

Proof of Corollary 1

Equation (27) gives $\partial \nu_t^H / \partial T_t = \phi_O \xi_O T_t^{\xi_O - 1} H_{t-1}^{-\psi_O} > 0$. Thus predetermined organizational capital affects choices only through a finite marginal cost schedule. The first-order conditions (40)–(41) determine x_t and e_t jointly; there is no separate equality requiring $T_t = H_{t-1}$ and hence no differential restriction $dx_t + \omega_e de_t = 0$. Such a restriction emerges only in the limiting case in which the cost of deviations from a fixed throughput level becomes infinite. Therefore the smooth-throughput specification removes the hard-capacity restriction by construction. □

Proof of Proposition 2

The tax result follows from Proposition 1, since $F_\tau < 0$ and $F_e < 0$ imply $\partial e_t / \partial \tau_{r,t} < 0$. A higher expected depreciation rate lowers the continuation value in (46), because it reduces the term $\mathbb{E}_t[\text{disc}_{t,t+1}(1 - \delta_{m,t}) q_{t+1}^m]$. Hence $\widetilde{q}_t^m$ falls, and Proposition 1 implies that e_t falls. Productive capability building rises only if this reduction in market-building relaxes ν_t^H; then $X_\nu < 0$ implies x_t rises. □

Proof of Proposition 3

The result is a direct application of Proposition 1. An increase in $\mathcal{M}_t$ raises the left side of the market-building FOC. If the induced increase in ν_t^H is not large enough to offset it, the unique local solution has higher e_t. Productive capability building depends on the productive composite and the same throughput shadow value. If a disturbance raises both composites by more than the induced increase in ν_t^H, both intangible margins rise. If the customer-building composite rises by more than the productive-building composite, market-building rises by more and the customer-capital state responds more persistently. This is the benchmark demand response. Monetary easing also raises both composites on impact, but it does so indirectly through the nominal and demand blocks, thus its customer-capital response is smaller than the demand response. □

Proof of Proposition 4

A marginal increase in e_t raises future customer capital by $\eta_M \xi_e e_t^{\xi_e - 1}$. The decentralized firm values the private continuation payoff q_{t+1}^m, but it does not internalize that

higher customer capital lowers future demand elasticity. From Equations (21)–(23),

$$\frac{\partial \varepsilon_{t+1}}{\partial \bar{m}_{t+1}} = -\frac{\phi_m \exp(\phi_{zM} z^M_{t+1})}{d_{t+1}} < 0, \qquad \frac{\partial \mu^*_{t+1}}{\partial \varepsilon_{t+1}} = -\frac{1}{(\varepsilon_{t+1}-1)^2} < 0.$$

Thus $\partial \mu^*_{t+1}/\partial \bar{m}_{t+1} > 0$ whenever $\phi_m > 0$ and $\varepsilon_{t+1} > 1$. A higher desired markup increases the equivalent monopoly wedge

$$\tau^{\mu}_{t+1} \equiv \frac{\mu^*_{t+1}-1}{\mu^*_{t+1}}, \qquad \frac{\partial \tau^{\mu}_{t+1}}{\partial \mu^*_{t+1}} = \frac{1}{(\mu^*_{t+1})^2} > 0.$$

For a planner that values strategic-sector output at household marginal utility, the allocative marginal welfare loss from this wedge is

$$\lambda_{t+1} p_{S,t+1} Y_{S,t+1} \frac{1}{(\mu^*_{t+1})^2} \frac{\partial \mu^*_{t+1}}{\partial \bar{m}_{t+1}} > 0,$$

because $\lambda_{t+1} > 0$, $p_{S,t+1} Y_{S,t+1} > 0$, and the derivative above is positive. Sticky prices add a nonnegative term whenever the induced movement in desired markups increases the Rotemberg loss. The organizational channel adds a nonnegative term whenever the private return to customer acquisition draws scarce throughput away from productive capability relative to the planner's allocation. Denote these two marginal losses by $\mathcal{R}_{\mu,t+1} \geq 0$ and $\mathcal{H}_{t+1} \geq 0$. Then

$$\lambda^{\text{ext}}_t = \mathbb{E}_t \left[\text{disc}_{t,t+1}\, \eta_M \xi_e e_t^{\xi_e - 1} \left(\lambda_{t+1} p_{S,t+1} Y_{S,t+1} \frac{1}{(\mu^*_{t+1})^2} \frac{\partial \mu^*_{t+1}}{\partial \bar{m}_{t+1}} + \mathcal{R}_{\mu,t+1} + \mathcal{H}_{t+1} \right) \right] > 0.$$

The planner's market-building condition is therefore the decentralized condition plus a positive shadow cost of customer-capital accumulation. □

Proof of Corollary 2

Competition policy lowers $\mathcal{M}_t$ through the tax and through the durability of customer capital. That reduces the market-building response that monetary stabilization otherwise needs to offset. Conversely, monetary stabilization reduces demand-driven movements in $\mathcal{M}_t$, which lowers the marginal value of cyclical competition intervention. The two instruments therefore act as local complements around the cooperative allocation. □